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**Part 3: Physical layer;**
**Release 2**

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# Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Digital Enhanced Cordless Telecommunications (DECT).

The present document is part 3 of a multi-part deliverable covering the DECT-2020 New Radio (NR) technology. Full details of the entire series can be found in part 1 [1].

DECT-2020 NR is recognized in Recommendation ITU-R M.2150 [i.1] as a component RIT fulfilling the IMT-2020 requirements of the IMT-2020 use scenarios URLLC and mMTC. The Set of Radio Interface Technology (SRIT) called "DECT 5G SRIT" is involving 3GPP NR and DECT-2020 NR.

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# 1 Scope

The present document is one of the parts of the specification of the DECT-2020 New Radio (NR).

The present document specifies the Physical layer and interaction between PHY and MAC layer.

# 2 References

## 2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 103 636-1: "DECT-2020 New Radio (NR); Part 1: Overview; Release 2".

[2] ETSI TS 103 636-2: "DECT-2020 New Radio (NR); Part 2: Radio reception and transmission requirements; Release 2".

[3] ETSI TS 103 636-4: "DECT-2020 New Radio (NR); Part 4: MAC layer; Release 2".

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[i.1] Recommendation ITU-R M.2150: "Detailed specifications of the terrestrial radio interfaces of International Mobile Telecommunications-2020 (IMT-2020)".

# 3 Definition of terms, symbols and abbreviations

## 3.1 Terms

Void.

## 3.2 Symbols

For the purposes of the present document, the following symbols apply:

| | |
|---|---|
| $\forall$ | Mathematical notation for "for all" |
| $\bigwedge$ | Mathematical notation for "and" |
| $\lfloor x \rfloor$ | Mathematical notation for "floor of x" i.e. rounding towards zero |
| $\lceil x \rceil$ | Mathematical notation for "ceiling of x" i.e. rounding towards infinity |
| $\beta$ | Fourier transform scaling factor |
| $\mu$ | Subcarrier scaling factor |
| $\Delta_f^\mu$ | Subcarrier spacing for given subcarrier scaling factor |
| $f_s^{\mu,\beta}$ | Sample frequency |
| $k_{OCC}^\beta$ | Occupied subcarriers for given transform scaling factor |
| $B_{DFT}^{\mu,\beta}$ | Nominal bandwidth |
| $B_{TX}^{\mu,\beta}$ | Transmission bandwidth |
| $GI^\mu$ | Guard interval for given subcarrier scaling factor |
| $M_{symb}^{stream}$ | Number of modulated symbols in a spatial stream |
| $M_{symb}$ | Number of modulated symbols |
| $N_{CP}^\beta$ | Cyclic Prefix size for given transform scaling factor |
| $N_{symb}^{DF}$ | Number of symbols in Data Field |
| $N_{symb}^{(GI+STF)}$ | Number of symbols in Guard Interval and STF combined |
| $N_{DFT}^\beta$ | Discrete Fourier Transform size for given Fourier transform scaling factor |
| $N_{OCC}^\beta$ | Number of occupied subcarriers for given Fourier transform scaling factor |
| $N_{re}^{DRS}$ | Number of DRS resource elements |
| $N_{re}^{PCC}$ | Number of PCC resource elements |
| $N_{re}^{PDC}$ | Number of PDC resource elements |
| $N_{symb}^{SLOT}$ | Number OFDM symbols in a slot |
| $N_{subslot}^{SLOT}$ | Number of subslots in a slot |
| $N_{slot}^{FRAME}$ | Number of slots in a frame |
| $N_{symb}^{PACKET}$ | Number of OFDM symbols in a transmission packet |
| $N_{TX}$ | Number of transmission antennas |
| $N_{TX}^{eff}$ | Effective number of transmission antennas |
| $N_{TS}$ | Number of transmit streams |
| $N_{SS}$ | Number of spatial streams |
| $N_{bps}$ | Number of bits per symbol for given modulation |
| $N_{REP}^{STF}$ | Number of repetitions of sequence $S$ in Synchronization Training Field |
| $T_{frame}$ | Duration of a frame |
| $T_{slot}$ | Duration of a slot |
| $T_s^{\mu,\beta}$ | Sample time interval |
| $T_{symb}^\mu$ | Duration of OFDM symbol for given subcarrier scaling factor |

## 3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI TS 103 636-1 [1] and the following apply:

NOTE: An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in ETSI TS 103 636-1 [1].

| | |
|---|---|
| ARQ | Automatic Repeat reQuest |
| BPSK | Binary Phase Shift Keying |
| CP | Cyclic Prefix |
| CRC | Cyclic Redundancy Check |
| DC | Zero or DC Subcarrier |
| DECT | Digital Enhanced Cordless Telecommunications |
| DF | Data Field |
| DFT | Discrete Fourier Transform |

| | |
|---|---|
| DRS | Demodulation Reference Signal |
| FDMA | Frequency Division Multiple Access |
| GF | Galois Field |
| GI | Guard Interval |
| HARQ | Hybrid ARQ |
| MAC | Medium Access Control |
| OFDM | Orthogonal Frequency Division Multiplexing |
| PCC | Physical Control Channel |
| PCCC | Parallel Concatenated Convolutional Code |
| PDC | Physical Data Channel |
| PDU | Protocol Data Unit |
| PHY | PHYsical layer |
| QAM | Quadrature Amplitude Modulation |
| QPSK | Quadrature Phase Shift Keying |
| RD | Radio Device |
| SAP | Service Access Point |
| SS | Spatial Stream |
| STF | Synchronization Training Field |
| TDD | Time Division Duplex |
| TDMA | Time Division Multiple Access |
| TS | Transmit Stream |
| TX | Transmission |

# 4 Physical layer principles

## 4.1 General description of Physical layer

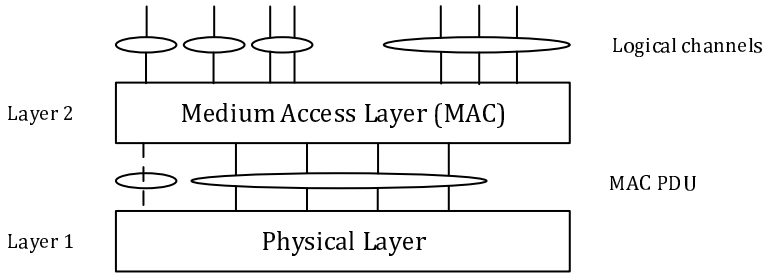



**Figure 4.1-1: Radio interface protocol architecture around the Physical layer**

Figure 4.1-1 shows the DECT-2020 NR radio interface protocol architecture around the Physical layer (PHY). The physical layer interfaces the Medium Access Control (MAC) layer. The circles between different layer/sub-layers indicate Service Access Points (SAPs). The physical layer offers Physical Control Channel (PCC) and Physical Data Channel (PDC) to transmit MAC PDU(s). Different physical channels are characterized by how the information is transferred over the radio interface within single transmission packet.

The physical layer performs the following functions in order to provide the data transport service:

- Error detection on the physical channels and indication to higher layers.
- FEC encoding/decoding of the physical channels.
- Hybrid ARQ soft-combining.
- Rate matching of the coded physical channel data to physical channels.
- Mapping of the coded physical channel data onto physical channels.
- Modulation and demodulation of physical channels.
- Frequency and time synchronization.

- Radio characteristics measurements and indication to higher layers.
- Multiple Input Multiple Output (MIMO) antenna processing.
- Transmit Diversity (TX diversity).
- Beamforming.

The physical channels defined are:

- the Physical Control Channel (PCC);
- the Physical Data Channel (PDC).

The modulation schemes supported are:

- BPSK;
- QPSK;
- 16-QAM;
- 64-QAM;
- 256-QAM; and
- 1024-QAM.

The channel coding in all of the physical channels is turbo coding with a rate 1/3 mother code punctured to the coderate of the channel or the selected MCS according to Table A-1. Trellis termination is used for the turbo coding. Before the turbo coding, transport blocks are segmented into byte aligned segments with a maximum codeblock size. Error detection is supported by the use of 16 or 24 bit CRC as specified for a given physical channel.

## 4.2 Multiple access

The multiple access scheme for the DECT-2020 NR physical layer is based on Time Division Duplex (TDD) combined with Frequency Division Multiple Access (FDMA) and Time Division Multiple Access (TDMA). The physical layer operates with non-overlapping channels in frequency domain and non-overlapping transmission slots in time domain. Radio channel spacing is defined in ETSI TS 103 636-2 [2].

The modulation within the transmitted packets is Orthogonal Frequency Division Multiplexing (OFDM) with a Cyclic Prefix (CP).

Both frame duration (10 ms) and slot duration (0,41667 ms) ensure coexistence with legacy DECT systems.

## 4.3 Numerologies

In the present document, unless otherwise noted, the size of various fields in the time domain are expressed in terms of basic parameters. Subcarrier spacing is defined by the subcarrier scaling factor $\mu$, resulting either in 27 kHz, 54 kHz, 108 kHz or 216 kHz OFDM subcarriers spacing $\Delta_f^{\mu}$. In addition, the Fourier transform scaling factor $\beta$ can be set to allow different transmission bandwidths for each configuration of the subcarrier spacing. The numerologies listed in Table 4.3-1 support multiple throughput and latency configurations for the network. In the table? $B_{DFT}^{\mu,\beta}$ denotes the nominal bandwidth, $B_{TX}^{\mu,\beta}$ denotes the transmission bandwidth consisting of $N_{OCC}^{\beta}$ occupied subcarriers and the empty DC carrier in the centre of the transmission bandwidth, $T_s^{\mu,\beta}$ denotes the critical sample rate, $N_{DFT}^{\beta}$ the Fourier transform size, $N_{CP}^{\beta}$ denotes the cyclic prefix size in samples.

**Table 4.3-1: Supported transmission numerologies**

| | | $\beta$ | $B_{DFT}^{\mu,\beta}$ [kHz] | $T_s^{\mu,\beta}$ | $N_{DFT}^{\beta}$ | $N_{CP}^{\beta}$ | $N_{OCC}^{\beta}$ | $B_{TX}^{\mu,\beta}$ [kHz] |
|---|---|---|---|---|---|---|---|---|
| $\mu$ | 1 | 1 | 1 728 | 5,7870E-07 | 64 | 8 | 56 | 1 539 |
| $\Delta_f^{\mu}$ [kHz] | 27 | 2 | 3 456 | 2,8935E-07 | 128 | 16 | 112 | 3 051 |
| $T_{symb}^{\mu}$ [us] | 41,667 | 4 | 6 912 | 1,4468E-07 | 256 | 32 | 224 | 6 075 |
| $N_{symb}^{SLOT,\mu}$ | 10 | 8 | 13 824 | 7,2338E-08 | 512 | 64 | 448 | 12 123 |
| $N_{subslot}^{SLOT,\mu}$ | 2 | 12 | 20 736 | 4,8225E-08 | 768 | 96 | 672 | 18 171 |
| $GI^{\mu}$ [us] | 18,52 | 16 | 27 648 | 3,6169E-08 | 1 024 | 128 | 896 | 24 219 |
| | | $\beta$ | $B_{DFT}^{\mu,\beta}$ [kHz] | $T_s^{\mu,\beta}$ | $N_{DFT}^{\beta}$ | $N_{CP}^{\beta}$ | $N_{OCC}^{\beta}$ | $B_{TX}^{\mu,\beta}$ [kHz] |
| $\mu$ | 2 | 1 | 3 456 | 2,8935E-07 | 64 | 8 | 56 | 3 078 |
| $\Delta_f^{\mu}$ [kHz] | 54 | 2 | 6 912 | 1,4468E-07 | 128 | 16 | 112 | 6 102 |
| $T_{symb}^{\mu}$ [us] | 20,833 | 4 | 13 824 | 7,2338E-08 | 256 | 32 | 224 | 12 150 |
| $N_{symb}^{SLOT,\mu}$ | 20 | 8 | 27 648 | 3,6169E-08 | 512 | 64 | 448 | 24 246 |
| $N_{subslot}^{SLOT,\mu}$ | 4 | 12 | 41 472 | 2,4113E-08 | 768 | 96 | 672 | 36 342 |
| $GI^{\mu}$ [us] | 20,83 | 16 | 55 296 | 1,8084E-08 | 1 024 | 128 | 896 | 48 438 |
| | | $\beta$ | $B_{DFT}^{\mu,\beta}$ [kHz] | $T_s^{\mu,\beta}$ | $N_{DFT}^{\beta}$ | $N_{CP}^{\beta}$ | $N_{OCC}^{\beta}$ | $B_{TX}^{\mu,\beta}$ [kHz] |
| $\mu$ | 4 | 1 | 6 912 | 1,4468E-07 | 64 | 8 | 56 | 6 156 |
| $\Delta_f^{\mu}$ [kHz] | 108 | 2 | 13 824 | 7,2338E-08 | 128 | 16 | 112 | 12 204 |
| $T_{symb}^{\mu}$ [us] | 10,417 | 4 | 27 648 | 3,6169E-08 | 256 | 32 | 224 | 24 300 |
| $N_{symb}^{SLOT,\mu}$ | 40 | 8 | 55 296 | 1,8084E-08 | 512 | 64 | 448 | 48 492 |
| $N_{subslot}^{SLOT,\mu}$ | 8 | 12 | 82 944 | 1,2056E-08 | 768 | 96 | 672 | 72 684 |
| $GI^{\mu}$ [us] | 10,42 | 16 | 110 592 | 9,0422E-09 | 1 024 | 128 | 896 | 96 876 |
| | | $\beta$ | $B_{DFT}^{\mu,\beta}$ [kHz] | $T_s^{\mu,\beta}$ | $N_{DFT}^{\beta}$ | $N_{CP}^{\beta}$ | $N_{OCC}^{\beta}$ | $B_{TX}^{\mu,\beta}$ [kHz] |
| $\mu$ | 8 | 1 | 13 824 | 7,2338E-08 | 64 | 8 | 56 | 12 312 |
| $\Delta_f^{\mu}$ [kHz] | 216 | 2 | 27 648 | 3,6169E-08 | 128 | 16 | 112 | 24 408 |
| $T_{symb}^{\mu}$ [us] | 5,208 | 4 | 55 296 | 1,8084E-08 | 256 | 32 | 224 | 48 600 |
| $N_{symb}^{SLOT,\mu}$ | 80 | 8 | 110 592 | 9,0422E-09 | 512 | 64 | 448 | 96 984 |
| $N_{subslot}^{SLOT,\mu}$ | 16 | 12 | 165 888 | 6,0282E-09 | 768 | 96 | 672 | 145 368 |
| $GI^{\mu}$ [us] | 10,42 | 16 | 221 184 | 4,5211E-09 | 1 024 | 128 | 896 | 193 752 |

## 4.4 Frame structure

The radio frame has a duration of $T_{frame} = 10\ ms$ and consists of $N_{slot}^{FRAME} = 24$ slots with a slot duration of $T_{slot} = 0{,}41667\ ms$ as depicted in Figure 4.4-1.

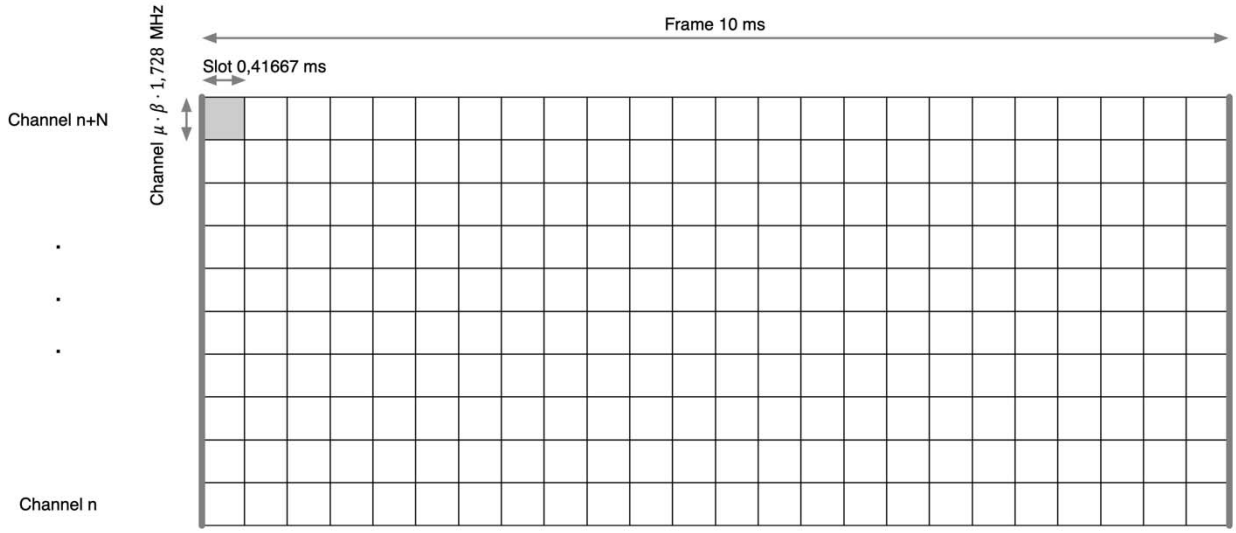



**Figure 4.4-1: DECT-2020 NR frame structure**

Each slot consists of $N_{symb}^{SLOT,\mu}$ = 10, 20, 40 or 80 OFDM symbols depending on subcarrier scaling factor $\mu$. Slot is further divided into $N_{subslot}^{SLOT,\mu}$ subslots according to Table 4.3-1 for each subcarrier scaling $\mu$. Packet transmission duration is integer multiple of subslots.

Basic channel width is 1,728 MHz. Multiple adjacent basic channels can be aggregated with $\beta$ and $\mu$ to form a wider transmission bandwidth ranging from 1,728 MHz to 221,184 MHz. Channel raster and numbering is specified in ETSI TS 103 636-2 [2].

## 4.5 Physical resources

Physical resources are mapped to resource elements $(s, k, l)_{\beta}$, where $s$ may denote either transmit stream or spatial stream index, $k$ denotes the subcarrier index and $l$ denotes the OFDM symbol position in the time domain relative to the start of the transmission packet as depicted in Figure 4.5-1. The occupied subcarriers indices are:

$$k_{OCC}^{\beta} = \left[-\frac{N_{OCC}^{\beta}}{2}, \cdots, -1, 1, \cdots, \frac{N_{OCC}^{\beta}}{2}\right],$$

where the $N_{OCC}^{\beta}$ is given by Table 4.3-1.The remaining subcarriers are the guard bands and the zero carrier (or DC carrier) which are not used for data transmission. Example of resource mappings are depicted in Figures 4.5-2 and 4.5-3.

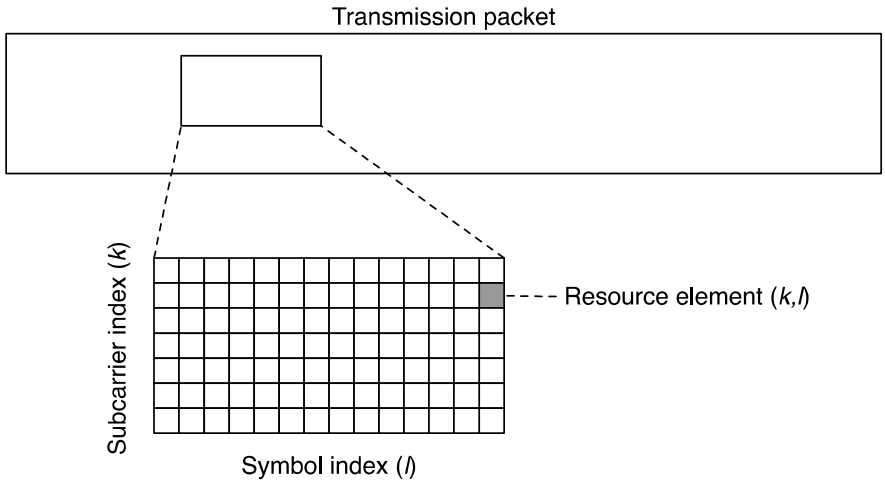



**Figure 4.5-1: Resource grid and indexing**

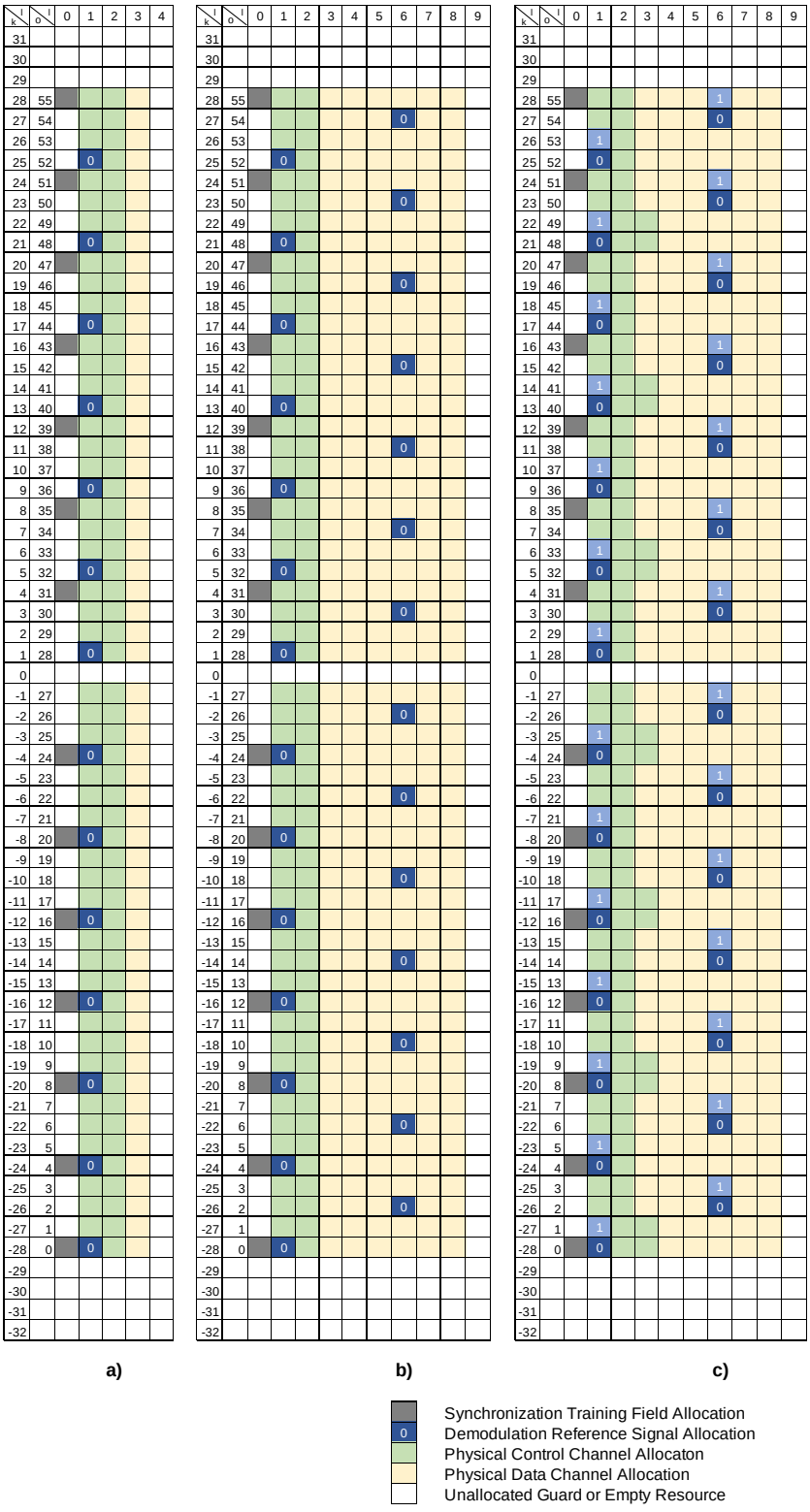



**Figure 4.5-2: Illustration of resource mapping for $(\mu, \beta) = (1, 1)$**
**a) Transmission from single effective antenna of one subslot duration**
**b) Transmission from single effective antenna of two subslots duration**
**c) Transmission from two effective antennas of two subslots duration**

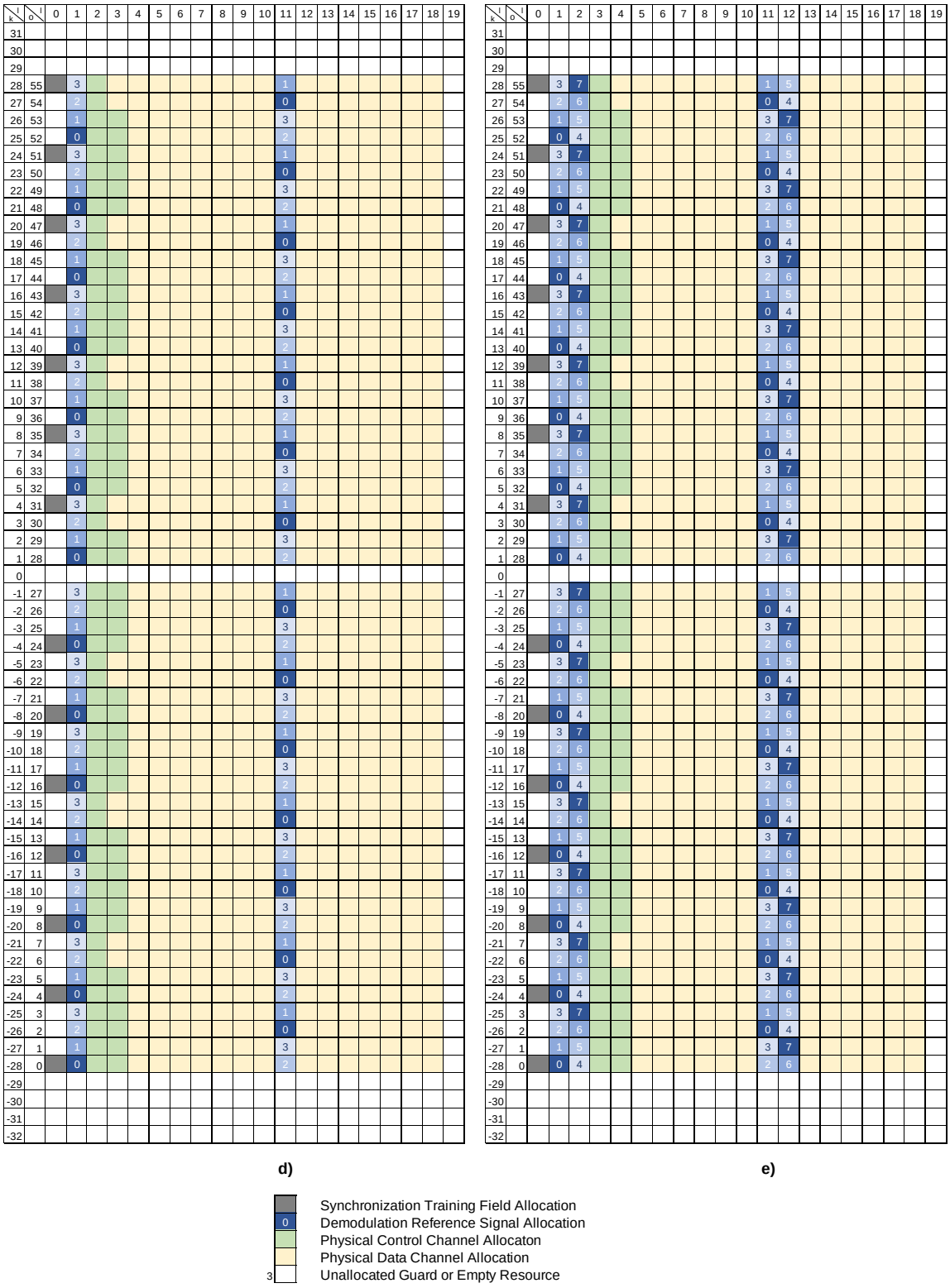



**Figure 4.5-3: Illustration of resource mapping for $(\mu, \beta) = (1, 1)$**
**d) Transmission from four effective antennas of four subslots duration**
**e) Transmission from eight effective antennas of four subslots duration**

# 5 Physical layer transmissions

## 5.1 Transmission packet structure

DECT-2020 NR transmission packet consists of Synchronization Training Field (STF), Data Field (DF) and Guard Interval (GI) as depicted in Figures 5.1-1, 5.1-2 and 5.1-3, where STF consists of 7 or 9 periodic repetitions of sequence S, and Data Field consists of a number of OFDM symbols D each with Cyclic Prefix CP. OFDM symbol length $T_{symb}^{\mu}$ is dependent on the subcarrier scaling factor $\mu$ as shown in Table 4.3-1. STF transmission starts at transmission allocation boundary. SFT is purposefully constructed to create time domain repetitive pattern for receiver gain, timing and frequency acquisition. DF carries Demodulation Reference Signal (DRS), Physical Control Channel (PCC) and Physical Data Channel (PDC). GI in the end of the packet allows transmission-reception and reception-transmission turnaround and to avoid overlapping transmissions from adjacent TDMA timeslots.

Transmission packet length in OFDM symbols is:

$$\text{if } PacketLengthType = 0 \quad \Rightarrow N_{symb}^{PACKET} = PacketLength * N_{symb}^{SLOT,\mu} / N_{subslot}^{SLOT,\mu}$$
$$\text{if } PacketLengthType = 1 \quad \Rightarrow N_{symb}^{PACKET} = PacketLength * N_{symb}^{SLOT,\mu}$$

depending whether *PacketLengthType* in Physical Header ETSI TS 103 636-4 [3], clause 6.2.1 indicates that the packet length is specified in terms of slots or subslots. The transmission packet length contains GI duration.

For $N_{TX}^{eff} \geq 4$ transmission length should be at least three subslots (15 OFDM symbols) to accomodate second set of demodulation reference signals for time variant channel and frequency error estimation.

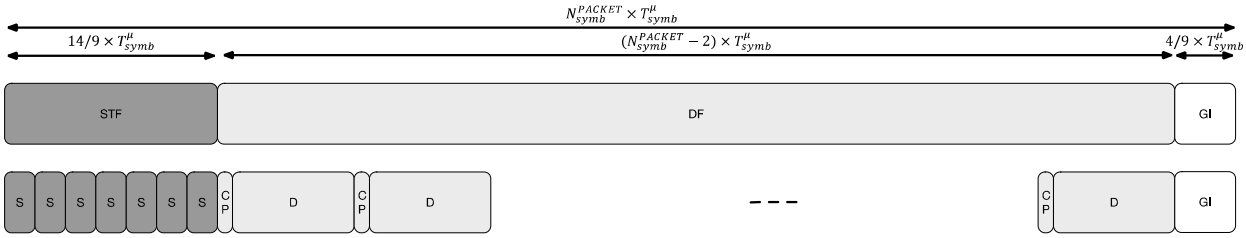



**Figure 5.1-1: Packet structure for $\mu = \{1\}$**

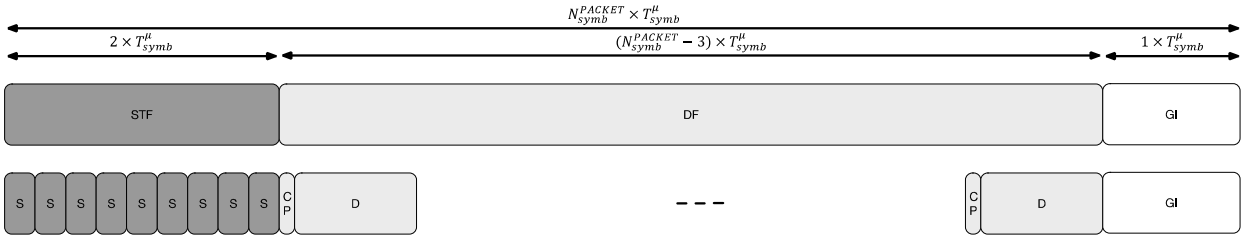



**Figure 5.1-2: Packet structure for $\mu = \{2, 4\}$**

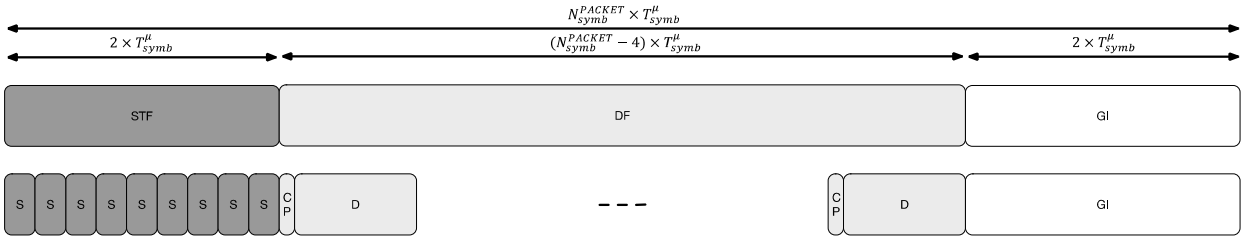



**Figure 5.1-3: Packet structure for $\mu = \{8\}$**

NOTE: For the highest subcarrier scaling the 4-bit packet length specifier allows packet length scaling from 5 to 80 OFDM symbols when packet length is specified in subslots, and up to 1 280 OFDM symbols when length is specified in slots.

# 5.2 Physical resource mapping

## 5.2.1 Guard Interval (GI)

GI in the end of the packet allows transmission-reception and reception-transmission turnaround and to avoid overlapping transmissions from adjacent timeslots. Guard intervals are of length $GI^{\mu=1} = \frac{4}{9} \times T^{\mu}_{symb}$, $GI^{\mu=\{2,4\}} = 1 \times T^{\mu}_{symb}$ and $GI^{\mu=8} = 2 \times T^{\mu}_{symb}$ for subcarrier scaling factors $\mu = \{1,2,4,8\}$, respectively. Guard interval duration in $\mu s$ is listed in Table 4.3-1 for each subcarrier scaling.

## 5.2.2 Synchronization Training Field (STF)

Synchronization training signal is mapped into resource elements:

$$(t, k_i, l) = \left(0, k^{\beta}_{OCC}[i \times 4], 0\right), \forall\, i = 0, \ldots, \frac{N^{\beta}_{OCC}}{8} - 1$$

and

$$(t, k_i, l) = \left(0, k^{\beta}_{OCC}\left[\frac{N^{\beta}_{OCC}}{2} + 3 + (i - \frac{N^{\beta}_{OCC}}{8}) \times 4\right], 0\right), \forall\, i = \frac{N^{\beta}_{OCC}}{8}, \ldots, \frac{N^{\beta}_{OCC}}{4} - 1$$

Thus, synchronization training is always in transmit stream $t = 0$ and in OFDM symbol $l = 0$ on every fourth subcarrier starting from the lowest occupied negative subcarrier but excluding the DC carrier.

EXAMPLE: The occupied STF subcarriers for DFT size 64 are [-28, -24, -20, -16, -12, -8, -4, 4, 8, 12, 16, 20, 24, 28] as depicted in Figure 4.5-2.

Define $N = length\left(y^{(0),\beta}\right)$ and $y^{(0),\beta}_r(k) = (-1)^{k+1} y^{(0),\beta}(N - 1 - k)$ for all $k = 0, \ldots, N - 1$ which flips the vector elements and alternates sign of every other element. The STF sequences are defined as:

$$y^{(0),\beta=1} = e^{j\pi/4}\{1, -1, 1, 1, -1, 1, 1, -1, 1, 1, 1, -1, -1, -1\}$$

$$y^{(0),\beta=2} = e^{j\pi/4}\{-1, 1, -1, 1, 1, -1, 1, 1, -1, 1, 1, 1, -1, 1, -1, -1, -1, 1, -1, -1, -1, 1, 1, 1, -1, -1, -1, -1\}$$

$$y^{(0),\beta=4} = e^{j\pi/4}\{-1, -1, -1, 1, -1, 1, -1, -1, 1, 1, 1, 1, -1, 1, -1, -1, -1, 1, -1, 1, 1, -1, -1, -1, -1, -1, 1, -1, 1, 1, 1, -1, 1, -1, 1, 1, -1, -1, -1, -1, 1, -1, -1, -1, -1, 1, -1, 1, 1, -1, -1, -1, -1, 1, -1\}$$

$$y^{(0),\beta=8} = \left\{y^{(0),\beta=4};\ y^{(0),\beta=4}_r\right\}$$

$$y^{(0),\beta=16} = \left\{y^{(0),\beta=8};\ y^{(0),\beta=8}_r\right\}$$

$$y^{(0),\beta=12}[\mathrm{i}] = y^{(0),\beta=16}[i + 2 \times 14], i = 0, 1, \ldots, 12 \times 14 - 1$$

## 5.2.3 Demodulation Reference Signal (DRS)

Demodulation Reference Signals are allocated to the transmit streams according to the number of transmit streams $N_{TS}$. DRS is transmitted on the resource elements:

$$(t, k_i, l) = \left(t, k_{OCC}^{\beta}[i \times 4 + (t + (n \bmod 2) \times 2) \bmod 4], 1 + \lfloor t/4 \rfloor + n \times N_{step}\right),$$
$$i = 0, \ldots, \frac{N_{OCC}^{\beta}}{4} - 1\ , n = 0, \ldots, \left\lfloor \frac{N_{symb}^{PACKET}}{N_{step}} \right\rfloor - 1$$

$$N_{step} = \begin{cases} 5, if\ N_{TX}^{eff} \leq 2 \\ 10, if\ N_{TX}^{eff} \geq 4 \end{cases},$$

where $t$ is the transmit stream index.

EXAMPLE: Thus the pilot carriers for DFT size of 64 and for $t = 0$ are [-28, -24, -20, -16, -12, -8, -4, 1, 5, 9, 13, 17, 21, 25], for OFDM symbols $1 + n \cdot N_{step} \forall\ n\ mod\ 2 = 0$ and [-26, -22, -18, -14, -10, -6, -2, 3, 7, 11, 15, 19, 23, 27] for $1 + n \cdot N_{step} \forall\ n\ mod\ 2 = 1$ as depicted in Figure 4.5-2.

Signal transmitted on DRS subcarrier $k_i$ is:

$$y_i^{DRS,(t)} = \begin{cases} y^{\beta}[4 \times i + \mathrm{t} \bmod 4]\ \forall\ i = 0, \ldots, N_{OCC}^{\beta}/4 - 1 \wedge \mathrm{t} < 4 \\ -y^{\beta}[4 \times i + \mathrm{t} \bmod 4]\ \forall\ i = 0, \ldots, N_{OCC}^{\beta}/4 - 1 \wedge \mathrm{t} \geq 4 \end{cases},$$

where the base sequences are defined as:

$$y^{\beta=1} = \{1, 1, 1, 1, -1, 1, 1, -1, -1, 1, 1, 1, 1, -1, 1, -1, 1, 1, -1, 1, -1, 1, -1, 1, 1, 1, 1, 1, -1, 1,$$
$$-1, -1, 1, 1, -1, -1, -1, -1, 1, -1, -1, -1, -1, -1, 1, 1, 1, -1, 1, 1, -1, -1, 1, -1, -1, -1\}$$

$$y^{\beta=2} = \left\{y^{\beta=1}, y^{\beta=1}\right\}$$

$$y^{\beta=4} = \left\{y^{\beta=2}, y^{\beta=2}\right\}$$

$$y^{\beta=8} = \left\{y^{\beta=4}, y^{\beta=4}\right\}$$

$$y^{\beta=12} = \left\{y^{\beta=4}, y^{\beta=4}, y^{\beta=4}\right\}$$

$$y^{\beta=16} = \left\{y^{\beta=8}, y^{\beta=8}\right\}$$

## 5.2.4 Physical Control Channel (PCC)

PCC is mapped to spatial stream 0 to the $N_{re}^{PCC} = 98$ resource elements starting from OFDM symbol $l = 1$ and to the resource elements which are not already occupied by DRS in any transmit stream. The procedure for resource element allocation for PCC is defined with steps:

1) Start from OFDM symbol $l = 1$ and set $N_{re}^{unalloc} = 98$.

2) Starting from the lowest subcarrier of the OFDM symbol $l$, select the subcarriers which are not allocated for DRS and denote them by $k_{(0,l)}, k_{(1,l)}, \ldots, k_{(U-1,l)}$, where $U$ is the number of such unoccupied carriers.

3) If $U < N_{re}^{unalloc}$ go to 4) else go to 5) to spread the remaining allocation as widely as possible across the transmission bandwidth.

4) Allocate all the available subcarriers in symbol $l$ to PCC:

   a) Add all subcarriers $k_{(0,l)}, k_{(1,l)}, \ldots, k_{(U-1,l)}$ to the set of resource elements $k_l^{PCC}$ allocated to PCC.

   b) Proceed to the next OFDM symbol by setting $l = l + 1$ and subtracting already allocated resource elements $N_{re}^{unalloc} = N_{re}^{unalloc} - U$.

   c) Jump to 2).

5) Assign $R^{PCC} = 7$ to be the number of rows of the matrix. The rows of the matrix are numbered 0, 1, 2,…, $R^{PCC} - 1$ from top to bottom.

6) Determine the number of columns of the matrix $C^{PCC}$ by:

$$C^{PCC} = U/R^{PCC}$$

The columns of rectangular matrix are numbered 0, 1, 2, … ,$C^{PCC} - 1$ from left to right.

7) Then, the subcarrier indices are written into the$(R^{PCC} \times C^{PCC})$ matrix row by row starting with bit $k_{(0,l)}$ in column 0 of row 0:

$$\begin{bmatrix} k_{(0,l)} & k_{(1,l)} & k_{(2,l)} & \cdots & k_{(C^{PCC}-1,l)} \\ k_{(C^{PCC},l)} & k_{(C^{PCC}+1,l)} & k_{(C^{PCC}+2,l)} & \cdots & k_{(2C^{PCC},l)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ k_{((R^{PCC}-1)\times C^{PCC},l)} & k_{((R^{PCC}-1)\times C^{PCC}+1,l)} & k_{((R^{PCC}-1)\times C^{PCC}+2,l)} & \cdots & k_{(R^{PCC}\times C^{PCC}-1,l)} \end{bmatrix}$$

8) The $N_{re}^{unalloc}$ subcarriers are read from the matrix column by column starting from row 0 of column 0 to the set of PCC resource elements $k_l^{PCC}$. Thus, allocation order is $k_{(0,l)}, k_{(C^{PCC},l)}, \ldots, k_{((R^{PCC}-1)\times C^{PCC},l)}, k_{(1,l)}, k_{(C^{PCC}+1,l)}, \ldots$, until all $N_{re}^{PCC}$ are allocated.

EXAMPLE: Examples of resource element allocations for PCC are depicted in Figures 4.5-2 and 4.5-3.

Modulated data is mapped to the set of resource elements $k^{PCC}$ starting from lowest OFDM symbol index $l$ and from lowest subcarrier index available for that OFDM symbol, filling the subcarriers first in frequency direction and then proceeding to the next OFDM symbol.

NOTE: Symbol mapping order is independent of allocation order. Thus, the list of allocated subcarriers is used in sorted order from lowest to highest for each OFDM symbol.

## 5.2.5 Physical Data Channel (PDC)

The remaining resource elements of DF which are not allocated for DRS in any transmit stream or PCC in spatial stream 0 are allocated for PDC.

The length of DF in OFDM symbols is given by:

$$N_{symb}^{DF} = N_{symb}^{PACKET} - N_{symb}^{GI+STF}$$

where $N_{symb}^{GI+STF} = 2$ for $\mu = \{1\}$, $N_{symb}^{GI+STF} = 3$ for $\mu = \{2, 4\}$ and $N_{symb}^{GI+STF} = 4$ for $\mu = 8$.

The number of DRS resource elements $N_{re}^{DRS}$ in a packet is given by:

$$N_{re}^{DRS} = N_{TX}^{eff} \times \frac{N_{OCC}^{\beta}}{4} \times \left\lfloor \frac{N_{symb}^{PACKET}}{N_{step}} \right\rfloor$$

where:

$$N_{step} = \begin{cases} 5, if\ N_{TX}^{eff} \leq 2 \\ 10, if\ N_{TX}^{eff} \geq 4 \end{cases}$$

The number of PDC resource elements $N_{re}^{PDC}$ is then given by:

$$N_{re}^{PDC} = N_{symb}^{DF} \times N_{OCC}^{\beta} - N_{re}^{DRS} - N_{re}^{PCC}$$

Modulated data is mapped to the set of resource elements starting from lowest OFDM symbol index $l$ and from lowest subcarrier index available for that OFDM symbol, filling the subcarriers first in frequency direction and then proceeding to the next OFDM symbol.

## 5.3 Transport block size

With number of resource elements $N_{re}^{PDC}$ available for transmission transport block size $N_{bits}^{TB}$ is calculated as follows.

Supported Modulation and Coding Schemes (MCSs) are defined in Table A-1. For a MCS carrying $N_{bps}$ bits per symbol and coding rate *R*, maximum number of bits which can be carried by the PDC is given by:

$$N_{bits}^{PDC} = \lfloor N_{ss} \times N_{re}^{PDC} \times N_{bps} \times R \rfloor,$$

where the number of resource elements available for PDC transmission $N_{re}^{PDC}$ for given packet size is calculated according to clause 5.2.5 and $N_{SS}$ is the number of parallel spatial streams.

Set the CRC length of the TBS and individual code block segments as:

$$L = 24$$

Set maximum turbo encoder code block size to:

$$Z = 2\ 048$$

If $N_{bits}^{PDC} \leq 512$, set:

$$M = 8$$

Else if $N_{bits}^{PDC} \leq 1\ 024$, set:

$$M = 16$$

Else if $N_{bits}^{PDC} \leq 2\ 048$, set:

$$M = 32$$

Else:

$$M = 64$$

Calculate the largest multiple of M not greater than $N_{bits}^{PDC}$ as:

$$N_M = \left\lfloor \frac{N_{bits}^{PDC}}{M} \right\rfloor \times M$$

If $N_M \leq Z$, set:

$$N_{bits}^{TB} = N_M - L$$

Else transport block will be segmented. Calculate the number of code block segments:

$$C = \left\lceil \frac{N_M - L}{Z} \right\rceil$$

To get the transport block size subtract the transport block CRC length and individual codeblock CRC lengths:

$$N_{bits}^{TB} = N_M - (C + 1) \times L$$

NOTE 1: With this definition of transport block size the number of filler bits in clause 6.1.3 is always 0.

NOTE 2: Configurations where β = 1, MCS = 0, packet length = 0, packet length type = 0 (i.e. one subslot) the resulting $N_{bits}^{TB}$ = 0. In these cases, the transmission is done leaving all PDC resources empty.

# 6 Generic procedures

## 6.1 Channel coding, rate-matching and interleaving

### 6.1.1 Overview

Data and control streams from/to MAC layer are encoded/decoded to offer physical layer packet services over the radio transmission link. Channel coding scheme is a combination of error detection, error correction, rate matching and interleaving.

### 6.1.2 CRC calculation

Denote the input bits to the CRC computation by $a_0, a_1, a_2, a_3, \ldots, a_{A-1}$, and the parity bits by $p_0, p_1, p_2, p_3, \ldots, p_{L-1}$. $A$ is the size of the input sequence and $L$ is the number of parity bits. The parity bits are generated by one of the following cyclic generator polynomials:

- $g_{CRC24A}(D) = [D^{24} + D^{23} + D^{18} + D^{17} + D^{14} + D^{11} + D^{10} + D^7 + D^6 + D^5 + D^4 + D^3 + D + 1]$; and
- $g_{CRC24B}(D) = [D^{24} + D^{23} + D^6 + D^5 + D + 1]$ for a CRC length $L = 24$; and
- $g_{CRC16}(D) = [D^{16} + D^{12} + D^5 + 1]$ for a CRC length $L = 16$.

The encoding is performed in a systematic form, which means that in GF(2), the polynomial:

$$a_0 D^{A+23} + a_1 D^{A+22} + \ldots + a_{A-1} D^{24} + p_0 D^{23} + p_1 D^{22} + \ldots + p_{22} D^1 + p_{23}$$

yields a remainder equal to 0 when divided by the corresponding length-24 CRC generator polynomial, $g_{CRC24A}(D)$ or $g_{CRC24B}(D)$, the polynomial:

$$a_0 D^{A+15} + a_1 D^{A+14} + \ldots + a_{A-1} D^{16} + p_0 D^{15} + p_1 D^{14} + \ldots + p_{14} D^1 + p_{15}$$

yields a remainder equal to 0 when divided by $g_{CRC16}(D)$.

The bits after CRC attachment are denoted by $b_0, b_1, b_2, b_3, \ldots, b_{B-1}$, where $B = A + L$. The relation between $a_k$ and $b_k$ is:

$b_k = a_k$ for $k = 0, 1, 2, \ldots, A-1$

$b_k = p_{k-A}$ for $k = A, A+1, A+2, \ldots, A+L-1$.

### 6.1.3 Code block segmentation

The input bit sequence to the code block segmentation is denoted by $b_0, b_1, b_2, b_3, \ldots, b_{B-1}$, where $B > 0$. If $B$ is larger than the maximum code block size $Z$, segmentation of the input bit sequence is performed, and an additional CRC sequence of $L = 24$ bits is attached to each code block. The maximum code block size is:

- $Z = 2\,048$.

If the number of filler bits $F$ calculated below is not 0, filler bits are added to the beginning of the first block.

Note that if $B < 40$, filler bits are added to the beginning of the code block.

The filler bits shall be set to <*NULL*> at the input to the encoder.

Total number of code blocks $C$ is determined by:

if $B \leq Z$

$L = 0$

Number of code blocks: $C = 1$

$$B' = B$$

else

$L$ = 24

Number of code blocks: $C = \lceil B/(Z - L) \rceil$.

$$B' = B + C \times L$$

end if

The bits output from code block segmentation, for $C \neq 0$, are denoted by $c_{r0}, c_{r1}, c_{r2}, c_{r3}, \ldots, c_{r(K_r-1)}$, where $r$ is the code block number, and $K_r$ is the number of bits for the code block number $r$.

Number of bits in each code block (applicable for $C \neq 0$ only):

First segmentation size: $K_+$= minimum $K$ in Table 6.1.4.2.3-1 such that $C \cdot K \geq B'$

if $C = 1$

the number of code blocks with length $K_+$ is $C_+$=1, $K_- = 0$, $C_- = 0$

else if $C > 1$

Second segmentation size: $K_-$= maximum $K$ in Table 6.1.4.2.3-1 such that $K < K_+$

$$\Delta_K = K_+ - K_-$$

Number of segments of size$K_-$: $C_- = \left\lfloor \frac{C \cdot K_+ - B'}{\Delta_K} \right\rfloor$

Number of segments of size$K_+$: $C_+ = C - C_-$

end if

Number of filler bits: $F = C_+ \times K_+ + C_- \times K_- - B'$

for $k$ = 0 to $F$-1 -- Insertion of filler bits

$$c_{0k} =< NULL >$$

end for

$k = F$

$s$ = 0

for $r$ = 0 to $C$-1

if $r < C_-$

$$K_r = K_-$$

else

$$K_r = K_+$$

end if

while $k < K_r - L$

$$c_{rk} = b_s$$

$$k = k + 1$$

$$s = s + 1$$

end while

if $C > 1$

The sequence $c_{r0}, c_{r1}, c_{r2}, c_{r3}, \ldots, c_{r(K_r-L-1)}$ is used to calculate the CRC parity bits $p_{r0}, p_{r1}, p_{r2}, \ldots, p_{r(L-1)}$ according to clause 6.1.2 with the generator polynomial $g_{CRC24B}(D)$. For CRC calculation it is assumed that filler bits, if present, have the value 0.

while $k < K_r$

$$c_{rk} = p_{r(k+L-K_r)}$$

$$k = k + 1$$

end while

end if

$$k = 0$$

end for

## 6.1.4 Channel coding

### 6.1.4.1 Introduction

The bit sequence input for a given code block to channel coding is denoted by $c_0, c_1, c_2, c_3, \ldots, c_{K-1}$, where $K$ is the number of bits to encode. After encoding the bits are denoted by $d_0^{(i)}, d_1^{(i)}, d_2^{(i)}, d_3^{(i)}, \ldots, d_{D-1}^{(i)}$, where $D$ is the number of encoded bits per output stream and $i$ indexes the encoder output stream. The relation between $c_k$ and $d_k^{(i)}$ and between $K$ and $D$ is dependent on the channel coding scheme.

### 6.1.4.2 Turbo coding

#### 6.1.4.2.1 Turbo encoder

The scheme of turbo encoder is a Parallel Concatenated Convolutional Code (PCCC) with two 8-state constituent encoders and one turbo code internal interleaver. The coding rate of turbo encoder is 1/3. The structure of turbo encoder is illustrated in Figure 6.1.4.2.1-1.

The transfer function of the 8-state constituent code for the PCCC is:

$$G(D) = \left[1, \frac{g_1(D)}{g_0(D)}\right],$$

where:

$$g_0(D) = 1 + D^2 + D^3,$$

$$g_1(D) = 1 + D + D^3.$$

The initial value of the shift registers of the 8-state constituent encoders shall be all zeros when starting to encode the input bits.

The output from the turbo encoder is:

$$d_k^{(0)} = x_k$$

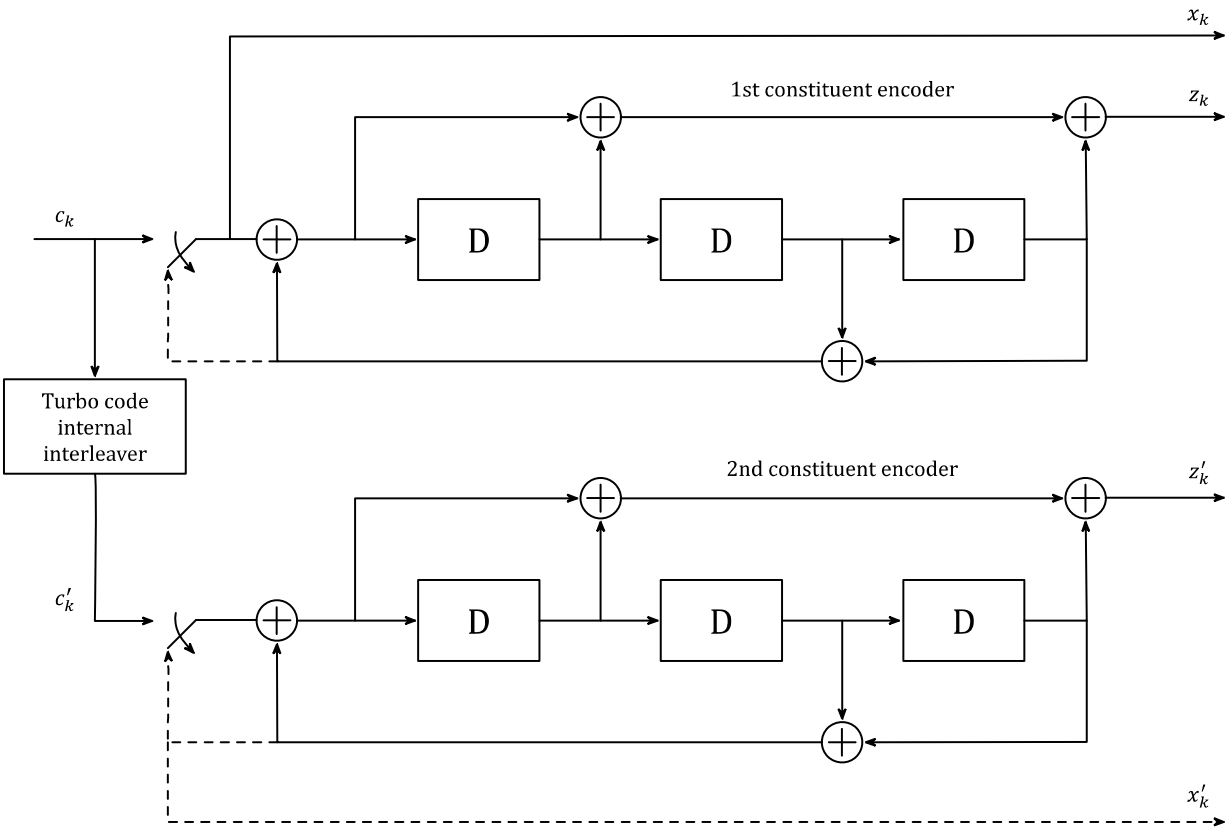
$$d_k^{(1)} = z_k$$

$$d_k^{(2)} = z'_k$$

for $k = 0,1,2,\ldots,K-1$.

If the code block to be encoded is the 0-th code block and the number of filler bits is greater than zero, i.e. $F > 0$, then the encoder shall set $c_k = 0$, $k = 0,\ldots,(F-1)$ at its input and shall set $d_k^{(0)} =< NULL >$, $k = 0,\ldots,(F-1)$ and $d_k^{(1)} =< NULL >$, $k = 0,\ldots,(F-1)$ at its output.

The bits input to the turbo encoder are denoted by $c_0, c_1, c_2, c_3, \ldots, c_{K-1}$, and the bits output from the first and second 8-state constituent encoders are denoted by $z_0, z_1, z_2, z_3, \ldots, z_{K-1}$ and $z'_0, z'_1, z'_2, z'_3, \ldots, z'_{K-1}$, respectively. The bits output from the turbo code internal interleaver are denoted by $c'_0, c'_1, \ldots, c'_{K-1}$, and these bits are to be the input to the second 8-state constituent encoder.




NOTE: Dotted lines apply for trellis termination only.

**Figure 6.1.4.2.1-1: Structure of rate 1/3 turbo encoder**

### 6.1.4.2.2 Trellis termination for turbo encoder

Trellis termination is performed by taking the tail bits from the shift register feedback after all information bits are encoded. Tail bits are padded after the encoding of information bits.

The first three tail bits shall be used to terminate the first constituent encoder (upper switch of Figure 6.1.4.2.1-1 in lower position) while the second constituent encoder is disabled. The last three tail bits shall be used to terminate the second constituent encoder (lower switch of Figure 6.1.4.2.1-1 in lower position) while the first constituent encoder is disabled.

The transmitted bits for trellis termination shall then be:

$d_K^{(0)} = x_K, d_{K+1}^{(0)} = z_{K+1}, d_{K+2}^{(0)} = x'_K, d_{K+3}^{(0)} = z'_{K+1}$

$d_K^{(1)} = z_K, d_{K+1}^{(1)} = x_{K+2}, d_{K+2}^{(1)} = z'_K, d_{K+3}^{(1)} = x'_{K+2}$

$d_K^{(2)} = x_{K+1}, d_{K+1}^{(2)} = z_{K+2}, d_{K+2}^{(2)} = x'_{K+1}, d_{K+3}^{(2)} = z'_{K+2}$

### 6.1.4.2.3 Turbo code internal interleaver

The bits input to the turbo code internal interleaver are denoted by $c_0, c_1, \ldots, c_{K-1}$, where $K$ is the number of input bits. The bits output from the turbo code internal interleaver are denoted by $c_0', c_1', \ldots, c_{K-1}'$.

The relationship between the input and output bits is as follows:

$$c_i' = c_{\Pi(i)}, i = 0, 1, \ldots, (K-1), i=0, 1, \ldots, (K-1)$$

where the relationship between the output index $i$ and the input index $\Pi(i)$ satisfies the following quadratic form:

$$\Pi(i) = (f_1 \cdot i + f_2 \cdot i^2) \bmod K$$

The parameters $f_1$ and $f_2$ depend on the block size $K$ and are summarized in Table 6.1.4.2.3-1.

**Table 6.1.4.2.3-1: Turbo code internal interleaver parameters**

| $i$ | $K$ | $f_1$ | $f_2$ | $i$ | $K$ | $f_1$ | $f_2$ | $i$ | $K$ | $f_1$ | $f_2$ | $i$ | $K$ | $f_1$ | $f_2$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 40 | 3 | 10 | 48 | 416 | 25 | 52 | 95 | 1 120 | 67 | 140 | 142 | 3 200 | 111 | 240 |
| 2 | 48 | 7 | 12 | 49 | 424 | 51 | 106 | 96 | 1 152 | 35 | 72 | 143 | 3 264 | 443 | 204 |
| 3 | 56 | 19 | 42 | 50 | 432 | 47 | 72 | 97 | 1 184 | 19 | 74 | 144 | 3 328 | 51 | 104 |
| 4 | 64 | 7 | 16 | 51 | 440 | 91 | 110 | 98 | 1 216 | 39 | 76 | 145 | 3 392 | 51 | 212 |
| 5 | 72 | 7 | 18 | 52 | 448 | 29 | 168 | 99 | 1 248 | 19 | 78 | 146 | 3 456 | 451 | 192 |
| 6 | 80 | 11 | 20 | 53 | 456 | 29 | 114 | 100 | 1 280 | 199 | 240 | 147 | 3 520 | 257 | 220 |
| 7 | 88 | 5 | 22 | 54 | 464 | 247 | 58 | 101 | 1 312 | 21 | 82 | 148 | 3 584 | 57 | 336 |
| 8 | 96 | 11 | 24 | 55 | 472 | 29 | 118 | 102 | 1 344 | 211 | 252 | 149 | 3 648 | 313 | 228 |
| 9 | 104 | 7 | 26 | 56 | 480 | 89 | 180 | 103 | 1 376 | 21 | 86 | 150 | 3 712 | 271 | 232 |
| 10 | 112 | 41 | 84 | 57 | 488 | 91 | 122 | 104 | 1 408 | 43 | 88 | 151 | 3 776 | 179 | 236 |
| 11 | 120 | 103 | 90 | 58 | 496 | 157 | 62 | 105 | 1 440 | 149 | 60 | 152 | 3 840 | 331 | 120 |
| 12 | 128 | 15 | 32 | 59 | 504 | 55 | 84 | 106 | 1 472 | 45 | 92 | 153 | 3 904 | 363 | 244 |
| 13 | 136 | 9 | 34 | 60 | 512 | 31 | 64 | 107 | 1 504 | 49 | 846 | 154 | 3 968 | 375 | 248 |
| 14 | 144 | 17 | 108 | 61 | 528 | 17 | 66 | 108 | 1 536 | 71 | 48 | 155 | 4 032 | 127 | 168 |
| 15 | 152 | 9 | 38 | 62 | 544 | 35 | 68 | 109 | 1 568 | 13 | 28 | 156 | 4 096 | 31 | 64 |
| 16 | 160 | 21 | 120 | 63 | 560 | 227 | 420 | 110 | 1 600 | 17 | 80 | 157 | 4 160 | 33 | 130 |
| 17 | 168 | 101 | 84 | 64 | 576 | 65 | 96 | 111 | 1 632 | 25 | 102 | 158 | 4 224 | 43 | 264 |
| 18 | 176 | 21 | 44 | 65 | 592 | 19 | 74 | 112 | 1 664 | 183 | 104 | 159 | 4 288 | 33 | 134 |
| 19 | 184 | 57 | 46 | 66 | 608 | 37 | 76 | 113 | 1 696 | 55 | 954 | 160 | 4 352 | 477 | 408 |
| 20 | 192 | 23 | 48 | 67 | 624 | 41 | 234 | 114 | 1 728 | 127 | 96 | 161 | 4 416 | 35 | 138 |
| 21 | 200 | 13 | 50 | 68 | 640 | 39 | 80 | 115 | 1 760 | 27 | 110 | 162 | 4 480 | 233 | 280 |
| 22 | 208 | 27 | 52 | 69 | 656 | 185 | 82 | 116 | 1 792 | 29 | 112 | 163 | 4 544 | 357 | 142 |
| 23 | 216 | 11 | 36 | 70 | 672 | 43 | 252 | 117 | 1 824 | 29 | 114 | 164 | 4 608 | 337 | 480 |
| 24 | 224 | 27 | 56 | 71 | 688 | 21 | 86 | 118 | 1 856 | 57 | 116 | 165 | 4 672 | 37 | 146 |
| 25 | 232 | 85 | 58 | 72 | 704 | 155 | 44 | 119 | 1 888 | 45 | 354 | 166 | 4 736 | 71 | 444 |
| 26 | 240 | 29 | 60 | 73 | 720 | 79 | 120 | 120 | 1 920 | 31 | 120 | 167 | 4 800 | 71 | 120 |
| 27 | 248 | 33 | 62 | 74 | 736 | 139 | 92 | 121 | 1 952 | 59 | 610 | 168 | 4 864 | 37 | 152 |
| 28 | 256 | 15 | 32 | 75 | 752 | 23 | 94 | 122 | 1 984 | 185 | 124 | 169 | 4 928 | 39 | 462 |
| 29 | 264 | 17 | 198 | 76 | 768 | 217 | 48 | 123 | 2 016 | 113 | 420 | 170 | 4 992 | 127 | 234 |
| 30 | 272 | 33 | 68 | 77 | 784 | 25 | 98 | 124 | 2 048 | 31 | 64 | 171 | 5 056 | 39 | 158 |
| 31 | 280 | 103 | 210 | 78 | 800 | 17 | 80 | 125 | 2 112 | 17 | 66 | 172 | 5 120 | 39 | 80 |
| 32 | 288 | 19 | 36 | 79 | 816 | 127 | 102 | 126 | 2 176 | 171 | 136 | 173 | 5 184 | 31 | 96 |
| 33 | 296 | 19 | 74 | 80 | 832 | 25 | 52 | 127 | 2 240 | 209 | 420 | 174 | 5 248 | 113 | 902 |
| 34 | 304 | 37 | 76 | 81 | 848 | 239 | 106 | 128 | 2 304 | 253 | 216 | 175 | 5 312 | 41 | 166 |
| 35 | 312 | 19 | 78 | 82 | 864 | 17 | 48 | 129 | 2 368 | 367 | 444 | 176 | 5 376 | 251 | 336 |
| 36 | 320 | 21 | 120 | 83 | 880 | 137 | 110 | 130 | 2 432 | 265 | 456 | 177 | 5 440 | 43 | 170 |
| 37 | 328 | 21 | 82 | 84 | 896 | 215 | 112 | 131 | 2 496 | 181 | 468 | 178 | 5 504 | 21 | 86 |
| 38 | 336 | 115 | 84 | 85 | 912 | 29 | 114 | 132 | 2 560 | 39 | 80 | 179 | 5 568 | 43 | 174 |
| 39 | 344 | 193 | 86 | 86 | 928 | 15 | 58 | 133 | 2 624 | 27 | 164 | 180 | 5 632 | 45 | 176 |
| 40 | 352 | 21 | 44 | 87 | 944 | 147 | 118 | 134 | 2 688 | 127 | 504 | 181 | 5 696 | 45 | 178 |
| 41 | 360 | 133 | 90 | 88 | 960 | 29 | 60 | 135 | 2 752 | 143 | 172 | 182 | 5 760 | 161 | 120 |
| 42 | 368 | 81 | 46 | 89 | 976 | 59 | 122 | 136 | 2 816 | 43 | 88 | 183 | 5 824 | 89 | 182 |
| 43 | 376 | 45 | 94 | 90 | 992 | 65 | 124 | 137 | 2 880 | 29 | 300 | 184 | 5 888 | 323 | 184 |
| 44 | 384 | 23 | 48 | 91 | 1 008 | 55 | 84 | 138 | 2 944 | 45 | 92 | 185 | 5 952 | 47 | 186 |
| 45 | 392 | 243 | 98 | 92 | 1 024 | 31 | 64 | 139 | 3 008 | 157 | 188 | 186 | 6 016 | 23 | 94 |
| 46 | 400 | 151 | 40 | 93 | 1 056 | 17 | 66 | 140 | 3 072 | 47 | 96 | 187 | 6 080 | 47 | 190 |
| 47 | 408 | 155 | 102 | 94 | 1 088 | 171 | 204 | 141 | 3 136 | 13 | 28 | 188 | 6 144 | 263 | 480 |

## 6.1.5 Rate matching

### 6.1.5.1 Rate matching for turbo coded transport channels

The rate matching for turbo coded transport channels is defined per coded block and consists of interleaving the three information bit streams $d_k^{(0)}$, $d_k^{(1)}$ and $d_k^{(2)}$, followed by the collection of bits and the generation of a circular buffer as depicted in Figure 6.1.5.1-1. The output bits for each code block are transmitted as described in clause 6.1.5.3.

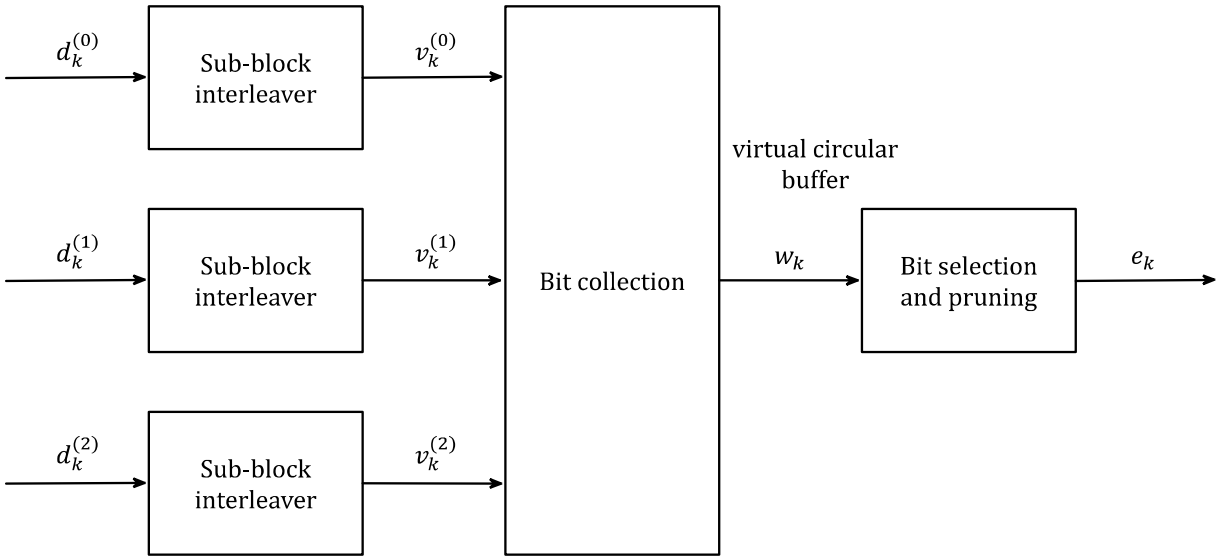



**Figure 6.1.5.1-1: Rate matching for turbo coded transport channels**

The bit stream $d_k^{(0)}$ is interleaved according to the sub-block interleaver defined in clause 6.1.5.2 with an output sequence defined as $v_0^{(0)}, v_1^{(0)}, v_2^{(0)}, \ldots, v_{K_\Pi-1}^{(0)}$and where $K_\Pi$ is defined in clause 6.1.5.2.

The bit stream $d_k^{(1)}$ is interleaved according to the sub-block interleaver defined in clause 6.1.5.2 with an output sequence defined as $v_0^{(1)}, v_1^{(1)}, v_2^{(1)}, \ldots, v_{K_\Pi-1}^{(1)}$.

The bit stream $d_k^{(2)}$ is interleaved according to the sub-block interleaver defined in clause 6.1.5.2 with an output sequence defined as $v_0^{(2)}, v_1^{(2)}, v_2^{(2)}, \ldots, v_{K_\Pi-1}^{(2)}$.

The sequence of bits $e_k$ for transmission is generated according to clause 6.1.5.3.

### 6.1.5.2 Sub-block interleaver

The bits input to the block interleaver are denoted by $d_0^{(i)}, d_1^{(i)}, d_2^{(i)}, \ldots, d_{D-1}^{(i)}$, where *D* is the number of bits. The output bit sequence from the block interleaver is derived as follows:

1) Assign $C_{subblock}^{TC} = 32$ to be the number of columns of the matrix. The columns of the matrix are numbered 0, 1, 2,…,$C_{subblock}^{TC} - 1$ from left to right.

2) Determine the number of rows of the matrix $R_{subblock}^{TC}$, by finding minimum integer $R_{subblock}^{TC}$ such that:

$$D \leq (R_{subblock}^{TC} \times C_{subblock}^{TC})$$

The rows of rectangular matrix are numbered 0, 1, 2, …,$R_{subblock}^{TC} - 1$ from top to bottom.

3) If $(R_{subblock}^{TC} \times C_{subblock}^{TC}) > D$, then $N_D = (R_{subblock}^{TC} \times C_{subblock}^{TC} - D)$ dummy bits are padded such that $y_k$ = <*NULL*> for $k$ = 0, 1,…, $N_D$ - 1. Then, $y_{N_D+k} = d_k^{(i)}$, $k$ = 0, 1,…, $D$-1, and the bit sequence $y_k$ is written into the $(R_{subblock}^{TC} \times C_{subblock}^{TC})$ matrix row by row starting with bit $y_0$ in column 0 of row 0:

$$\begin{bmatrix} y_0 & y_1 & y_2 & \cdots & y_{C_{subblock}^{TC}-1} \\ y_{C_{subblock}^{TC}} & y_{C_{subblock}^{TC}+1} & y_{C_{subblock}^{TC}+2} & \cdots & y_{2C_{subblock}^{TC}-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ y_{(R_{subblock}^{TC}-1)\times C_{subblock}^{TC}} & y_{(R_{subblock}^{TC}-1)\times C_{subblock}^{TC}+1} & y_{(R_{subblock}^{TC}-1)\times C_{subblock}^{TC}+2} & \cdots & y_{(R_{subblock}^{TC}\times C_{subblock}^{TC}-1)} \end{bmatrix}$$

For $d_k^{(0)}$ and $d_k^{(1)}$:

4) Perform the inter-column permutation for the matrix based on the pattern $\langle P(j)\rangle_{j\in\{0,1,\ldots,C_{subblock}^{TC}-1\}}$ that is shown in Table 6.1.5.2-1, where P($j$) is the original column position of the $j$-th permuted column. After permutation of the columns, the inter-column permuted $(R_{subblock}^{TC} \times C_{subblock}^{TC})$ matrix is equal to:

$$\begin{bmatrix} y_{P(0)} & y_{P(1)} & y_{P(2)} & \cdots & y_{P(C_{subblock}^{TC}-1)} \\ y_{P(0)+C_{subblock}^{TC}} & y_{P(1)+C_{subblock}^{TC}} & y_{P(2)+C_{subblock}^{TC}} & \cdots & y_{P(C_{subblock}^{TC}-1)+C_{subblock}^{TC}} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ y_{P(0)+(R_{subblock}^{TC}-1)\times C_{subblock}^{TC}} & y_{P(1)+(R_{subblock}^{TC}-1)\times C_{subblock}^{TC}} & y_{P(2)+(R_{subblock}^{TC}-1)\times C_{subblock}^{TC}} & \cdots & y_{P(C_{subblock}^{TC}-1)+(R_{subblock}^{TC}-1)\times C_{subblock}^{TC}} \end{bmatrix}$$

5) The output of the block interleaver is the bit sequence read out column by column from the inter-column permuted $(R_{subblock}^{TC} \times C_{subblock}^{TC})$ matrix. The bits after sub-block interleaving are denoted by:

$v_0^{(i)}, v_1^{(i)}, v_2^{(i)}, \ldots, v_{K_\Pi-1}^{(i)}$, where $v_0^{(i)}$ corresponds to $y_{P(0)}$, $v_1^{(i)}$ to $y_{P(0)+C_{subblock}^{TC}}$… and $K_\Pi = (R_{subblock}^{TC} \times C_{subblock}^{TC})$.

For $d_k^{(2)}$:

6) The output of the sub-block interleaver is denoted by $v_0^{(2)}, v_1^{(2)}, v_2^{(2)}, \ldots, v_{K_\Pi-1}^{(2)}$, where $v_k^{(2)} = y_{\pi(k)}$ and where:

$$\pi(k) \rightleftarrows = \left(P\left(\left\lfloor \frac{k}{R_{subblock}^{TC}} \right\rfloor\right) + C_{subblock}^{TC} \times (k \bmod R_{subblock}^{TC}) + 1\right) \bmod K_\Pi$$

The permutation function $P$ is defined in Table 6.1.5.2-1.

**Table 6.1.5.2-1: Inter-column permutation pattern for sub-block interleaver**

| Number of columns $C_{subblock}^{TC}$ | Inter-column permutation pattern $< P(0), P(1), \ldots, P(C_{subblock}^{TC}-1) >$ |
|---|---|
| 32 | < 0, 16, 8, 24, 4, 20, 12, 28, 2, 18, 10, 26, 6, 22, 14, 30, 1, 17, 9, 25, 5, 21, 13, 29, 3, 19, 11, 27, 7, 23, 15, 31 > |

### 6.1.5.3 Bit collection, selection and transmission

The circular buffer of length $K_w = 3K_\Pi$ for the *r*-th coded block is generated as follows:

$w_k = v_k^{(0)}$ for $k = 0,\ldots, K_\Pi - 1$

$w_{K_\Pi+2k} = v_k^{(1)}$ for $k = 0,\ldots, K_\Pi - 1$

$w_{K_\Pi+2k+1} = v_k^{(2)}$ for $k = 0,\ldots, K_\Pi - 1$

Denote the soft buffer size for the transport block by $N_{\text{IR}}$ bits and the soft buffer size for the *r*-th code block by $N_{cb}$ bits. The size $N_{cb}$ is obtained as follows, where *C* is the number of code blocks computed in clause 6.1.3:

$$N_{cb} = min\left(\left\lfloor\frac{N_{IR}}{C}\right\rfloor, K_w\right),$$

where $N_{\text{IR}}$ is equal to:

$$N_{IR} = \left\lfloor\frac{N_{soft}}{min\left(M_{\text{DL_HARQ}}, M_{\text{limit}}\right)}\right\rfloor$$

Where $N_{\text{soft}}$ is the total number of soft channel bits according to the Radio Device class category defined in Annex B.

$M_{\text{DL_HARQ}}$ is the maximum number of DL HARQ processes according to the Radio Device class category defined in Annex B.

$M_{\text{limit}}$ is a constant equal to 8.

Denoting by *E* the rate matching output sequence length for the *r*-th coded block, and $rv_{idx}$ the redundancy version number for this transmission ($rv_{idx}$ = 0, 1, 2 or 3), the rate matching output bit sequence is $e_k$, $k = 0,1,\ldots, E - 1$.

Define by $G = N_{re}^{PDC} \times N_{SS} \times N_{bps}$ the total number of bits available for the transmission of one transport block.

Set $G' = G/\left(N_{SS} \times N_{bps}\right) = N_{re}^{PDC}$ where $N_{bps}$ is equal to 1 for BPSK, 2 for QPSK, 4 for 16-QAM, 6 for 64-QAM, 8 for 256-QAM and 10 for 1024-QAM, and where:

$N_{SS}$ is equal to the number of spatial layers a transport block is mapped onto.

Set $\gamma = G' \bmod C$, where *C* is the number of code blocks computed in clause 6.1.3.

if $r \leq C - \gamma - 1$

set $E = N_{SS} \times N_{bps} \times \lfloor G'/C \rfloor = N_{SS} \times N_{bps} \times \lfloor N_{re}^{PDC}/C \rfloor$

else

set $E = N_{SS} \times N_{bps} \times \lceil G'/C \rceil = N_{SS} \times N_{bps} \times \lceil N_{re}^{PDC}/C \rceil$

end if

Set $k_0 = R_{subblock}^{TC} \times \left(2 \times \left\lfloor\frac{N_{cb}}{8 \cdot R_{subblock}^{TC}}\right\rfloor \times rv_{idx} + 2\right)$, where $R_{subblock}^{TC}$ is the number of rows defined in clause 6.1.5.2.

Set *k* = 0 and *j* = 0

while $\{k < E\}$

if $w_{(k_0+j) \bmod N_{cb}} \neq < NULL >$

$$e_k = w_{(k_0+j) \bmod N_{cb}}$$

$k = k + 1$

end if

$j = j + 1$

end while

### 6.1.6 Code block concatenation

The input bit sequence for the code block concatenation block are the sequences $e_{rk}$, for $r = 0, \ldots, C-1$ and $k = 0, \ldots, E_r - 1$. The output bit sequence from the code block concatenation block is the sequence $f_k$ for $k = 0, \ldots, G-1$.

The code block concatenation consists of sequentially concatenating the rate matching outputs for the different code blocks. Therefore:

set $k = 0$ and $r = 0$

while $r < C$

Set $j = 0$

while $j < E_r$

$$f_k = e_{rj}$$

$$k = k + 1$$

$$j = j + 1$$

end while

$$r = r + 1$$

end while

## 6.2 Pseudo-random sequence generation

Generic pseudo random sequence is defined by a length 31 Gold sequence. The output sequence $g(n)$ if length $N_{PN}$ where $n = 0,1, \cdots, N_{PN} - 1$ is defined by:

$$\begin{aligned} g(n) &= \left(x_1(n + N_C) + x_2(n + N_C)\right) mod\ 2 \\ x_1(n+31) &= \left(x_1(n+3) + x_1(n)\right) mod\ 2 \\ x_2(n+31) &= \left(x_2(n+3) + x_2(n+2) + x_2(n+1) + x_2(n)\right) mod\ 2 \end{aligned},$$

where $N_C = 1\ 600$ and the first m-sequence shall be initialized with:

$$x_1(0) = 1, x_1(n) = 0, \forall\, n = \{1,2, \cdots, 30\}$$

The initialization of the second m-sequence $x_2(n)$ is denoted by:

$$g_{init} = \sum_{i=0}^{30} x_2(i) \times 2^i,$$

with the value depending on the application sequence.

# 6.3 Modulation

## 6.3.1 Symbol mapping

### 6.3.1.1 Overview

The modulation mapper takes binary digits, 0 or 1, as input and produces complex-valued modulation symbols as output.

### 6.3.1.2 BPSK

In case of BPSK modulation, bit $d(i)$ is mapped to complex-valued modulation symbol $x(i)$ according to:

$$x(i) = \frac{1}{\sqrt{2}}\{(1 - 2 \times d(i)) + j(1 - 2 \times d(i))\}$$

### 6.3.1.3 QPSK

In case of QPSK modulation, pair of bits, $d(i),\ d(i+1)$ are mapped to complex-valued modulation symbols $x(i)$ according to:

$$x(i) = \frac{1}{\sqrt{2}}\{(1 - 2 \times d(i)) + j(1 - 2 \times d(i+1))\}$$

### 6.3.1.4 16-QAM

In case of 16-QAM modulation, quadruplets of bits, $d(i),\ d(i+1),\ d(i+2),\ d(i+3)$ are mapped to complex-valued modulation symbols $x(i)$ according to:

$$x(i) = \frac{1}{\sqrt{10}}\Big\{(1 - 2 \times d(i))\Big(2 - \big(1 - 2 \times d(i+2)\big)\Big) + j\,(1 - 2 \times d(i+1))\Big(2 - \big(1 - 2 \times d(i+3)\big)\Big)\Big\}$$

### 6.3.1.5 64-QAM

In case of 64-QAM modulation, hextuplets of bits, $d(i),\ d(i+1),\ d(i+2),\ d(i+3),\ d(i+4),\ d(i+5)$, are mapped to complex-valued modulation symbols $x(i)$ according to:

$$x(i) = \frac{1}{\sqrt{42}}\Big\{(1 - 2 \times d(i))\Big(4 - \big(1 - 2 \times d(i+2)\big)\Big(2 - \big(1 - 2 \times d(i+4)\big)\Big)\Big) + j\,(1 - 2 \times d(i+1))\Big(4 - \big(1 - 2 \times d(i+3)\big)\Big(2 - \big(1 - 2 \times d(i+5)\big)\Big)\Big)\Big\}$$

### 6.3.1.6 256-QAM

In case of 256-QAM modulation, octuplets of bits, $d(i),\ d(i+1),\ d(i+2),\ d(i+3),\ d(i+4),\ d(i+5),\ d(i+6),\ d(i+7)$ are mapped to complex-valued modulation symbols $x(i)$ according to:

$$x(i) = \frac{1}{\sqrt{170}}\Big\{(1 - 2 \times d(i))\Big(8 - \big(1 - 2 \times d(i+2)\big)\Big(4 - \big(1 - 2 \times d(i+4)\big)\Big(2 - \big(1 - 2 \times d(i+6)\big)\Big)\Big)\Big) + j\,(1 - 2 \times d(i+1))\Big(8 - \big(1 - 2 \times d(i+3)\big)\Big(4 - \big(1 - 2 \times d(i+5)\big)\Big(2 - \big(1 - 2 \times d(i+7)\big)\Big)\Big)\Big)\Big\}$$

#### 6.3.1.7 1024-QAM

In case of 1024-QAM modulation, decuplets of bits, $d(i)$, $d(i+1)$, $d(i+2)$, $d(i+3)$, $d(i+4)$, $d(i+5)$, $d(i+6)$, $d(i+7)$, $d(i+8)$, $d(i+9)$ are mapped to complex-valued modulation symbols $x(i)$ according to:

$$x(i) = \frac{1}{\sqrt{682}}\Big\{(1-2\times d(i))\Big(16-(1-2\times d(i+2))\Big(8-(1-2\times d(i+4))\Big(4-(1-2\times d(i+6))\Big(2-(1-2\times d(i+8))\Big)\Big)\Big)\Big)$$
$$+j(1-2\times d(i+1))\Big(16-(1-2\times d(i+3))\Big(8-(1-2\times d(i+5))\Big(4-(1-2\times d(i+7))\Big(2-(1-2\times d(i+9))\Big)\Big)\Big)\Big)\Big\}$$

### 6.3.2 Spatial multiplexing

The modulated transport block to spatial stream multiplexing shall be done according to Table 6.3.2-1, where the complex valued modulation symbols $x(i), i = 0,\ldots,M_{symb}$ is mapped to the modulated symbol in spatial stream $x^{(s)}(i), i = 0,\ldots,M_{symb}^{stream}-1$ for stream $s$ and $M_{symb}^{stream} = M_{symb}/N_{SS}$. The number of spatial streams is less than or equal to the number of antenna ports used for the transmission of the channel.

**Table 6.3.2-1: Modulated transport block to spatial stream mapping**

| Number of spatial streams | Modulated transport block to spatial stream mapping |
|---|---|
| 1 | $x^{(0)}(i) = x(i)$ |
| 2 | $x^{(0)}(i) = x(2i)$<br>$x^{(1)}(i) = x(2i+1)$ |
| 4 | $x^{(0)}(i) = x(4i)$<br>$x^{(1)}(i) = x(4i+1)$<br>$x^{(2)}(i) = x(4i+2)$<br>$x^{(3)}(i) = x(4i+3)$ |
| 8 | $x^{(0)}(i) = x(8i)$<br>$x^{(1)}(i) = x(8i+1)$<br>$x^{(2)}(i) = x(8i+2)$<br>$x^{(3)}(i) = x(8i+3)$<br>$x^{(4)}(i) = x(8i+4)$<br>$x^{(5)}(i) = x(8i+5)$<br>$x^{(6)}(i) = x(8i+6)$<br>$x^{(7)}(i) = x(8i+7)$ |

### 6.3.3 Transmit stream mapping

#### 6.3.3.1 Transmit stream mapping for spatial multiplexing and for single antenna

If transmit diversity is not used the precoding matrix is identity matrix, thus:

$$y^{(s)}(i) = x^{(s)}(i), i = 0,..,M_{symb}^{stream}-1, s = 0,\ldots,N_{SS}$$

where $y^{(s)}(i)$ denotes transmission in transmit stream index $s$ and $x^{(s)}(i)$ is modulated symbol for spatial stream index $s$ and time index $i$ and number of transmit streams $N_{TS} = N_{SS}$.

### 6.3.3.2 Transmit diversity precoding

When transmit diversity is used the pair of symbols $\left[x^{(0)}(2i), x^{(0)}(2i+1)\right]^T, i = 0,..,M_{symb}^{stream}/2 - 1$ of spatial stream 0 shall be precoded according to:

$$\begin{bmatrix} y^{(0)}(2i) \\ \vdots \\ y^{(N_{TS}-1)}(2i) \\ y^{(0)}(2i+1) \\ \vdots \\ y^{(N_{TS}-1)}(2i+1) \end{bmatrix} = Y_i \begin{bmatrix} Re\{x^{(0)}(2i)\} \\ Re\{x^{(0)}(2i+1)\} \\ Im\{x^{(0)}(2i)\} \\ Im\{x^{(0)}(2i+1)\} \end{bmatrix}, i = 0,..,M_{symb}^{stream} - 1$$

where $y^{(t)}(l)$ is the transmit diversity precoded transmission in transmit stream $t$ and index $l$ and $x^{(s)}(l)$ is the modulated symbol for spatial stream index $s$ and time index $l$. The precoding matrices $Y_i$ are given by Tables 6.3.3.2-1 to 6.3.3.2-3 for index $i$ and for each transmit diversity scheme.

**Table 6.3.3.2-1: Precoding matrix $Y_i$ for two antenna transmit diversity using two transmit streams**

| $Y_i$ |
|---|
| $\begin{bmatrix} 1 & 0 & j & 0 \\ 0 & -1 & 0 & j \\ 0 & 1 & 0 & j \\ 1 & 0 & -j & 0 \end{bmatrix}$ |

**Table 6.3.3.2-2: Precoding matrix $Y_i$ for four antenna transmit diversity using four transmit streams**

| index $i \bmod 6$ | $Y_i$ (ordered from left to right in increasing order of index) | | | |
|---|---|---|---|---|
| 0 - 3 | $\begin{bmatrix} 1 & 0 & j & 0 \\ 0 & -1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & j \\ 1 & 0 & -j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ | $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & j & 0 \\ 0 & -1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & j \\ 1 & 0 & -j & 0 \end{bmatrix}$ | $\begin{bmatrix} 1 & 0 & j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 1 & 0 & -j & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ | $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 1 & 0 & -j & 0 \end{bmatrix}$ |
| 4 - 5 | $\begin{bmatrix} 1 & 0 & j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & j \\ 0 & 1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & -j & 0 \end{bmatrix}$ | $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & j & 0 \\ 0 & -1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & j \\ 1 & 0 & -j & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ | - | - |

**Table 6.3.3.2-3: Precoding matrix $Y_i$ for eight antenna transmit diversity using eight transmit streams**

| index $i \bmod 12$ | $Y_i$ (ordered from left to right in increasing order of index) | | | |
|---|---|---|---|---|
| 0 - 3 | $\begin{bmatrix} 1 & 0 & j & 0 \\ 0 & -1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & j \\ 1 & 0 & -j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ | $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & j & 0 \\ 0 & -1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & j \\ 1 & 0 & -j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ | $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & j & 0 \\ 0 & -1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & j \\ 1 & 0 & -j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ | $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & j & 0 \\ 0 & -1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & j \\ 1 & 0 & -j & 0 \end{bmatrix}$ |
| 4 - 7 | $\begin{bmatrix} 1 & 0 & j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & -j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ | $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & -j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ | $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & -j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ | $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & -j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 8 - 11 | $\begin{bmatrix} 1 & 0 & j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 1 & 0 & -j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ | $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & -j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ | $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & -j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ | $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & j \\ 1 & 0 & -j & 0 \end{bmatrix}$ |

## 6.3.4 Beamforming and antenna port mapping

The block of vectors $[y^{(0)}(i) \quad \cdots \quad y^{(N_{TS}-1)}(i)]^T, i = 0,1,\ldots,M_{symb} - 1$ for $N_{TS}$ transmit streams shall be beamformed according to:

$$\begin{bmatrix} z^{(0)}(i) \\ \vdots \\ z^{(N_{TX}-1)}(i) \end{bmatrix} = W \begin{bmatrix} y^{(0)}(i) \\ \vdots \\ y^{(N_{TS}-1)}(i) \end{bmatrix}$$

where $z^{(p)}(i)$ is the beamformed transmission on antenna port $p$.

The same precoding or beamforming is applied to PCC, PDC, DRS and STF transmit streams for the precoding to be transparent to the receiver.

Single antenna transmission precoding matrix is identity matrix $W = 1$.

Open loop MIMO and transmit diversity transmissions can be done with any orthogonal precoding matrix which is transparent to the receiver.

The precoding matrix $W$ are given by Tables 6.3.4-1 to 6.3.4-5 for each closed loop codebook index applicable for given transmission.

Channel sounding packet shall always be transmitted with identity precoding matrix, thus using codebook index 0 of Tables 6.3.4-3, 6.3.4-5 and 6.3.4-6.

**Table 6.3.4-1: Precoding matrix $W$ for single transmit stream transmission using two antenna ports**

| Codebook index | $W$ (ordered from left to right in increasing order of index) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 - 5 | $\begin{bmatrix}1\\0\end{bmatrix}$ | $\begin{bmatrix}0\\1\end{bmatrix}$ | $\frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}$ | $\frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$ | $\frac{1}{\sqrt{2}}\begin{bmatrix}1\\j\end{bmatrix}$ | $\frac{1}{\sqrt{2}}\begin{bmatrix}1\\-j\end{bmatrix}$ | – | – |

**Table 6.3.4-2: Precoding matrix $W$ for single transmit stream transmission using four antenna ports**

| Codebook index | $W$ (ordered from left to right in increasing order of index) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 - 7 | $\begin{bmatrix}1\\0\\0\\0\end{bmatrix}$ | $\begin{bmatrix}0\\1\\0\\0\end{bmatrix}$ | $\begin{bmatrix}0\\0\\1\\0\end{bmatrix}$ | $\begin{bmatrix}0\\0\\0\\1\end{bmatrix}$ | $\frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\1\\0\end{bmatrix}$ | $\frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\-1\\0\end{bmatrix}$ | $\frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\j\\0\end{bmatrix}$ | $\frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\-j\\0\end{bmatrix}$ |
| 8 - 15 | $\frac{1}{\sqrt{2}}\begin{bmatrix}0\\1\\0\\1\end{bmatrix}$ | $\frac{1}{\sqrt{2}}\begin{bmatrix}0\\1\\0\\-1\end{bmatrix}$ | $\frac{1}{\sqrt{2}}\begin{bmatrix}0\\1\\0\\j\end{bmatrix}$ | $\frac{1}{\sqrt{2}}\begin{bmatrix}0\\1\\0\\-j\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1\\1\\1\\1\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1\\1\\j\\j\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1\\1\\-1\\-1\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1\\1\\-j\\-j\end{bmatrix}$ |
| 16 - 23 | $\frac{1}{2}\begin{bmatrix}1\\j\\1\\j\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1\\j\\j\\-1\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1\\j\\-1\\-j\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1\\j\\-j\\1\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1\\-1\\1\\-1\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1\\-1\\j\\-j\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1\\-1\\-1\\1\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1\\-1\\-j\\j\end{bmatrix}$ |
| 24 - 27 | $\frac{1}{2}\begin{bmatrix}1\\-j\\1\\-j\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1\\-j\\j\\1\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1\\-j\\-1\\j\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1\\-j\\-j\\-1\end{bmatrix}$ | – | – | – | – |

**Table 6.3.4-3: Precoding matrix $W$ for dual transmit stream transmission using two antenna ports**

| Codebook index | $W$ (ordered from left to right in increasing order of index) | | | |
|---|---|---|---|---|
| 0 - 2 | $\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1 & 1\\j & -j\end{bmatrix}$ | – |

**Table 6.3.4-4: Precoding matrix $W$ for dual transmit stream transmission using four antenna ports**

| Codebook index | $W$ (ordered from left to right in increasing order of index) | | | |
|---|---|---|---|---|
| 0 - 3 | $\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 0\\0 & 1\\0 & 0\\0 & 0\end{bmatrix}$ | $\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 0\\0 & 0\\0 & 1\\0 & 0\end{bmatrix}$ | $\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 0\\0 & 0\\0 & 0\\0 & 1\end{bmatrix}$ | $\frac{1}{\sqrt{2}}\begin{bmatrix}0 & 0\\1 & 0\\0 & 1\\0 & 0\end{bmatrix}$ |
| 4 - 7 | $\frac{1}{\sqrt{2}}\begin{bmatrix}0 & 0\\1 & 0\\0 & 0\\0 & 1\end{bmatrix}$ | $\frac{1}{\sqrt{2}}\begin{bmatrix}0 & 0\\0 & 0\\1 & 0\\0 & 1\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1 & 0\\0 & 1\\1 & 0\\0 & -j\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1 & 0\\0 & 1\\1 & 0\\0 & j\end{bmatrix}$ |
| 8 - 11 | $\frac{1}{2}\begin{bmatrix}1 & 0\\0 & 1\\-j & 0\\0 & 1\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1 & 0\\0 & 1\\-j & 0\\0 & -1\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1 & 0\\0 & 1\\-1 & 0\\0 & -j\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1 & 0\\0 & 1\\-1 & 0\\0 & j\end{bmatrix}$ |
| 12 - 15 | $\frac{1}{2}\begin{bmatrix}1 & 0\\0 & 1\\j & 0\\0 & 1\end{bmatrix}$ | $\frac{1}{2}\begin{bmatrix}1 & 0\\0 & 1\\j & 0\\0 & -1\end{bmatrix}$ | $\frac{1}{2\sqrt{2}}\begin{bmatrix}1 & 1\\1 & 1\\1 & -1\\1 & -1\end{bmatrix}$ | $\frac{1}{2\sqrt{2}}\begin{bmatrix}1 & 1\\1 & 1\\j & -j\\j & -j\end{bmatrix}$ |
| 16 - 19 | $\frac{1}{2\sqrt{2}}\begin{bmatrix}1 & 1\\j & j\\1 & -1\\j & -j\end{bmatrix}$ | $\frac{1}{2\sqrt{2}}\begin{bmatrix}1 & 1\\j & j\\j & -j\\-1 & 1\end{bmatrix}$ | $\frac{1}{2\sqrt{2}}\begin{bmatrix}1 & 1\\-1 & -1\\1 & -1\\-1 & 1\end{bmatrix}$ | $\frac{1}{2\sqrt{2}}\begin{bmatrix}1 & 1\\-1 & -1\\j & -j\\-j & j\end{bmatrix}$ |
| 20 - 21 | $\frac{1}{2\sqrt{2}}\begin{bmatrix}1 & 1\\-j & -j\\1 & -1\\-j & j\end{bmatrix}$ | $\frac{1}{2\sqrt{2}}\begin{bmatrix}1 & 1\\-j & -j\\j & -j\\1 & -1\end{bmatrix}$ | – | – |

**Table 6.3.4-5: Precoding matrix $W$ for four transmit stream transmission using four antenna ports**

| Codebook index | $W$ (ordered from left to right in increasing order of index) | | | |
|---|---|---|---|---|
| 0 - 3 | $\frac{1}{2}\begin{bmatrix}1 & 0 & 0 & 0\\0 & 1 & 0 & 0\\0 & 0 & 1 & 0\\0 & 0 & 0 & 1\end{bmatrix}$ | $\frac{1}{2\sqrt{2}}\begin{bmatrix}1 & 1 & 0 & 0\\0 & 0 & 1 & 1\\1 & -1 & 0 & 0\\0 & 0 & 1 & -1\end{bmatrix}$ | $\frac{1}{2\sqrt{2}}\begin{bmatrix}1 & 1 & 0 & 0\\0 & 0 & 1 & 1\\j & -j & 0 & 0\\0 & 0 & j & -j\end{bmatrix}$ | $\frac{1}{4}\begin{bmatrix}1 & 1 & 1 & 1\\1 & -1 & 1 & -1\\1 & 1 & -1 & -1\\1 & -1 & -1 & 1\end{bmatrix}$ |
| 4 | $\frac{1}{4}\begin{bmatrix}1 & 1 & 1 & 1\\1 & -1 & 1 & -1\\j & j & -j & -j\\j & -j & -j & j\end{bmatrix}$ | – | – | – |

**Table 6.3.4-6: Precoding matrix $W$ for eight transmit stream transmission using eight antenna ports**

| Codebook index | $W$ (ordered from left to right in increasing order of index) | |
|---|---|---|
| 0 | $\frac{1}{2\sqrt{2}}\begin{bmatrix}1 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\0 & 1 & 0 & 0 & 0 & 0 & 0 & 0\\0 & 0 & 1 & 0 & 0 & 0 & 0 & 0\\0 & 0 & 0 & 1 & 0 & 0 & 0 & 0\\0 & 0 & 0 & 0 & 1 & 0 & 0 & 0\\0 & 0 & 0 & 0 & 0 & 1 & 0 & 0\\0 & 0 & 0 & 0 & 0 & 0 & 1 & 0\\0 & 0 & 0 & 0 & 0 & 0 & 0 & 1\end{bmatrix}$ | – |

### 6.3.5 OFDM signal generation

$$q_{n,l}^{(p)} = \frac{1}{\sqrt{N_{OCC,l}}} \sum_k z_{k,l}^{(p)} \exp\left(\mathrm{j}\left(2\pi \times \frac{k}{N_{DFT}^{\beta}}\right) \times n\right), k = 0, \ldots, N_{DFT} - 1, n = 0, \ldots, N_{DFT} - 1$$

where $l$ is the OFDM symbol index, $N_{OCC,l}$ is number of occupied subcarriers for OFDM symbol $l$, $N_{DFT}^{\beta}$ is the discrete Fourier transform size for the scaling factor $\beta$, and $z_{k,l}^{(p)}$ is the frequency domain signal to be transformed and $q_{n,l}^{(p)}$ is the corresponding time domain signal for antenna port $p$. Sum of the transmit power over all transmit chains shall be normalized to 1,0 for each OFDM symbol $l$.

NOTE 1: Power scaling due to the transmit diversity or spatial multiplexing is built in to the precoding matrices.

NOTE 2: Number of occupied subcarriers is four times lower for STF symbol than for the other symbols, i.e. $4 \times N_{OCC,0} = N_{OCC,i}, i \neq 0$.

### 6.3.6 Cyclic prefix insertion

Cyclic prefixes for each of the Fourier transform scaling $\beta$ is 1/8 of the $N_{DFT}^{\beta}$ for all OFDM symbols except for the OFDM symbol 0 of the transmission packet.

For OFDM symbol 0 of the transmission packet the cyclic prefix shall equal to $3/4 \cdot N_{DFT}^{\beta}$ when subcarrier scaling factor $\mu = 1$ or $5/4 \cdot N_{DFT}^{\beta}$ when subcarrier scaling factor $\mu \in \{2, 4, 8\}$. In time domain, the OFDM symbol 0 of the transmission packet i.e. the STF signal, consists of $N_{REP}^{STF} = 7$ for $\mu = 1$ and $N_{REP}^{STF} = 9$ for $\mu \in \{2,4,8\}$ identical repetitions of the length $1/4 \cdot N_{DFT}^{\beta}$ base sequence.

### 6.3.7 STF signal cover sequence

Denote length $1/4 \cdot N_{DFT}^{\beta}$ base sequence by $S$. A following cover sequence shall be applied on the signal repetitions:

$$c_{\mu} = \{1, -1, 1, 1, -1, -1, -1\}, \mu = 1$$

$$c_{\mu} = \{1, -1, 1, 1, -1, -1, -1, -1, -1\}, \mu \in \{2,4,8\}.$$

The time domain STF signal is:

$$c_{\mu} \otimes S = \{c_{\mu}(0) \times S, \ldots, c_{\mu}(N_{REP}^{STF} - 1) \times S\},$$

which is Kronecker product of cover sequence with base sequence $S$.

Only for the purpose of transmitter conformance testing, RD may transmit STF without the cover sequence defined in this clause. Test cases where transmission without cover sequence can be enabled are indicated within the test case.

# 7 Transmission encoding

## 7.1 Transmitter block diagram

High level transmitter block diagram is depicted in Figure 7.1-1. $N_{SS}$ denotes the number of spatial streams. For PCC the number of spatial streams is fixed to one. The number of spatial streams for PDC is signalled within the Physical Header as specified ETSI TS 103 636-4 [3], clause 6.2.1. $N_{TX}^{eff}$ is the effective number of transmit antennas. Number of transmit streams $N_{TS}$ can be appropriately deduced from $N_{TX}^{eff}$ and $N_{SS}$ according to Table 7.2-1. The additional degrees of freedom beyond $N_{TX}^{eff}$ can be used for beamforming and the actual number of transmit antennas $N_{TX}$ does not need to be known to the receiver.

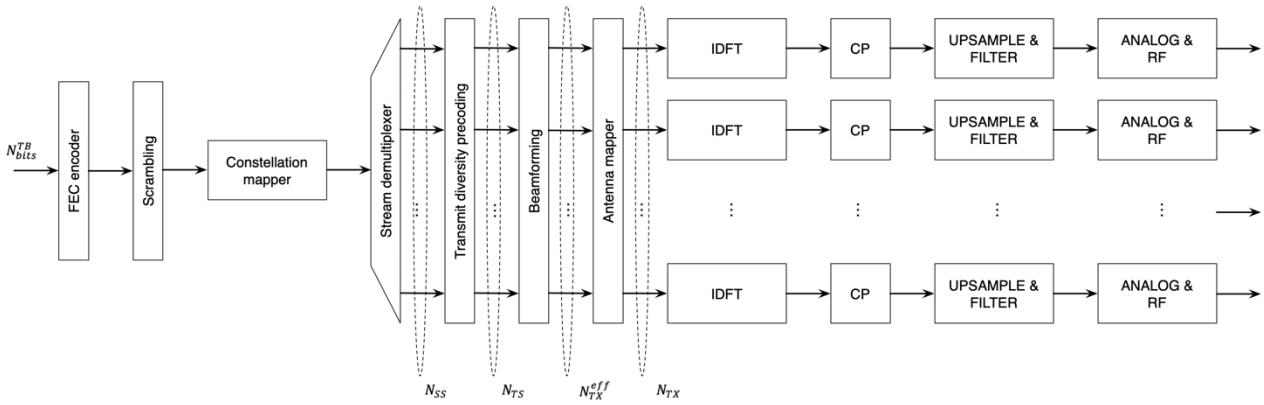



**Figure 7.1-1: Transmitter Block Diagram**

## 7.2 Transmission modes

Transmission modes available for transmitting a packet are listed in Table 7.2-1. The transmission mode is signalled with:

- Physical header field containing the number of spatial streams $N_{SS}$ signalled as specified in ETSI TS 103 636-4 [3].
- XOR-masking the CRC of the PCC as specified in clause 7.5.2.2 for closed loop transmission.
- XOR-masking the CRC of the PCC as specified in clause 7.5.2.3 for beamformed (precoded) transmission:
  - When precoding matrix is identity matrix the CRC of the PCC shall **NOT** be XOR-masked.
  - When precoding something else than identity matrix the CRC of the PCC shall be XOR-masked.

Closed loop transmission are beamformed according to clause 6.3.4 and beamforming matrix shall be uniformly applied to all fields of the transmission: STF, DRS, PCC and PDC.

Open loop transmissions can be beamformed according to clause 6.3.4 or with any other orthogonal beamforming matrix as long as the beamforming is transparent to the receiver and the beamforming is uniformly applied to all fields of transmission: STF, DRS, PCC and PDC.

NOTE: When transmission is not beamformed and it is signalled appropriately, receiver may use the channel estimates for channel sounding purposes. When transmission is beamformed the receiver sees composite of beamforming and channel response matrix.

**Table 7.2-1: Transmission modes and transmission mode signalling**
**CL - Closed Loop (True/False), BF - Beamformed (True/False), $N_{TX}$ - Number of antenna ports**

| **Radio device class $N_{SS}$ capability = 1 as defined in Annex B** | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | **PDC** | | | **PCC** | |
| **Transmission mode signalling** | $N_{TX}^{eff}$ | $N_{SS}$ | **CL** | **BF** | **Effective transmission mode** | $N_{TS}$ | $N_{TX}$ | **Effective transmission mode** | **BF** |
| Single antenna | 1 | 1 | F | F | Single antenna | 1 | 1 | Single antenna | F |
| **Radio device class $N_{SS}$ capability = 2 as defined in Annex B** | | | | | | | | | |
| | | | | | **PDC** | | | **PCC** | |
| **Transmission mode signalling** | $N_{TX}^{eff}$ | $N_{SS}$ | **CL** | **BF** | **Effective transmission mode** | $N_{TS}$ | $N_{TX}$ | **Effective transmission mode** | **BF** |
| Transmit diversity | 2 | 1 | F | T/F | 2 x 1 TXDiv | 2 | 2 | 2 x 1 TXDiv | T/F |
| MIMO open loop | 2 | 2 | F | T/F | 2 x 2 MIMO | 2 | 2 | 2 x 1 TXDiv | T/F |
| MIMO closed loop | 1 | 1 | T | T/F | Single antenna | 1 | 2 | Single antenna | T/F |
| MIMO closed loop | 2 | 2 | T | T/F | 2 x 2 MIMO | 2 | 2 | 2 x 1 TXDiv | T/F |
| **Radio device class $N_{SS}$ capability = 4 as defined in Annex B** | | | | | | | | | |
| | | | | | **PDC** | | | **PCC** | |
| **Transmission mode signalling** | $N_{TX}^{eff}$ | $N_{SS}$ | **CL** | **BF** | **Effective transmission mode** | $N_{TS}$ | $N_{TX}$ | **Effective transmission mode** | **BF** |
| Transmit diversity | 4 | 1 | F | T/F | 4 x 1 TXDiv | 4 | 4 | 4 x 1 TXDiv | T/F |
| MIMO open loop | 4 | 4 | F | T/F | 4 x 4 MIMO | 4 | 4 | 4 x 1 TXDiv | T/F |
| MIMO closed loop | 1 | 1 | T | T/F | Single antenna | 1 | 4 | Single antenna | T/F |
| MIMO closed loop | 2 | 2 | T | T/F | 2 x 2 MIMO | 2 | 4 | 2 x 1 TXDiv | T/F |
| MIMO closed loop | 4 | 4 | T | T/F | 4 x 4 MIMO | 4 | 4 | 4 x 1 TXDiv | T/F |
| **Radio device class $N_{SS}$ capability = 8 as defined in Annex B** | | | | | | | | | |
| | | | | | **PDC** | | | **PCC** | |
| **Transmission mode signalling** | $N_{TX}^{eff}$ | $N_{SS}$ | **CL** | **BF** | **Effective transmission mode** | $N_{TS}$ | $N_{TX}$ | **Effective transmission mode** | **BF** |
| Transmit diversity | 8 | 1 | F | T/F | 8 x 1 TXDiv | 8 | 8 | 8 x 1 TXDiv | T/F |
| MIMO open loop | 8 | 8 | F | T/F | 8 x 8 MIMO | 8 | 8 | 8 x 1 TXDiv | T/F |

## 7.3 Synchronization Training Field (STF) beamforming

Synchronization training field symbols $y_i^{STF,(0)}$ from transmit stream 0 as specified in clause 5.2.2 are beamformed with beamforming matrix defined in clause 6.3.4 for transmission as:

$$\begin{bmatrix} z^{(0)}(i) \\ \vdots \\ z^{(N_{TX}-1)}(i) \end{bmatrix} = W \begin{bmatrix} y^{STF,(0)}(i) \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

where the number of rows in the transmit stream vector is according to the number of transmit streams for the selected transmission mode.

# 7.4 Demodulation Reference Signal (DRS) beamforming

Demodulation reference symbols $y_i^{DRS,(t)}$ from transmit stream t as specified in clause 5.2.3 are beamformed with beamforming matrix defined in clause 6.3.4 for transmission as:

$$\begin{bmatrix} z^{(0)}(i) \\ \vdots \\ z^{(N_{TX}-1)}(i) \end{bmatrix} = W \begin{bmatrix} y^{DRS,(0)}(i) \\ \vdots \\ y^{DRS,(N_{TS}-1)}(i) \end{bmatrix},$$

where the number of rows in the transmit stream vector is according to the number of transmit streams for the selected transmission mode.

# 7.5 Physical Control Channel (PCC) encoding

## 7.5.1 Overall description

High level description of PCC encoding procedure is depicted in Figure 7.5.1-1. Number of physical control channel payload bits $N_{bits}^{PCC}$ is either 40 or 80 bits depending on the control channel format. The physical control channel is transmitted on $N_{re}^{PCC} = 98$ subcarriers. The receiver shall blind decode both transport block sizes and select the one with a CRC match.

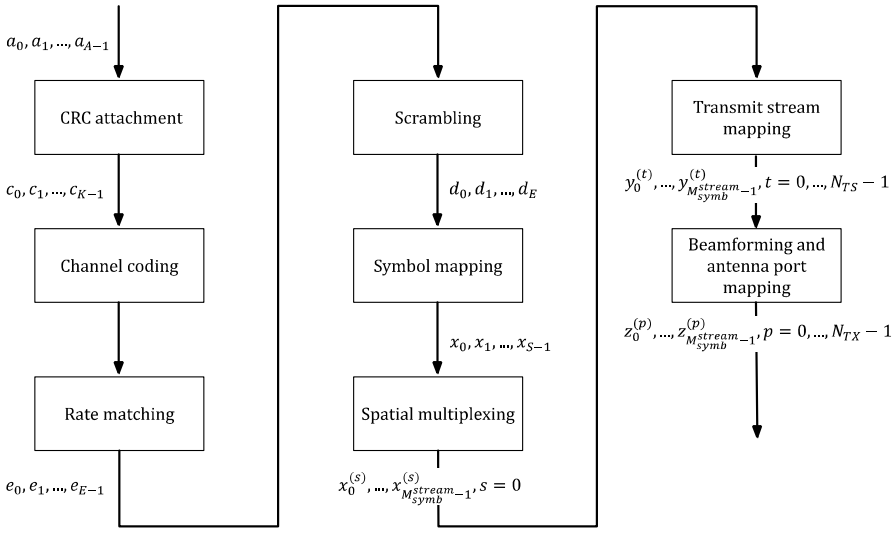



**Figure 7.5.1-1: Physical control channel encoding**

## 7.5.2 CRC calculation

### 7.5.2.1 Parity bit calculation

Physical control channel bits are $a_0, a_1, \cdots, a_{A-1}$, where $A = N_{bits}^{PCC}$ and the 16 Parity bits $p_0, p_1, \cdots, p_{15}$ for the PCC are generated by using polynome $g_{CRC16}(D)$ according to clause 6.1.2.

### 7.5.2.2 CRC Masking for MIMO closed loop

The 16 parity bits $p_0, p_1, \cdots, p_{15}$ are XOR masked with bitmask 0x5555 for signalling closed loop transmission.

### 7.5.2.3 CRC Masking for beamforming

The 16 parity bits $p_0, p_1, \cdots, p_{15}$ are XOR masked with bitmask 0xAAAA for signalling beamformed transmission.

#### 7.5.2.4 CRC attachment

CRC attachment is defined in clause 6.1.2. The bits after CRC attachment are denoted by $c_0, c_1, \ldots, c_{K-1}$, where $K = A + L$, where $L = 16$.

### 7.5.3 Channel coding & rate matching

The bit sequence $c_0, c_1, \ldots, c_{K-1}$ is channel encoded using the turbo encoder defined in clause 6.1.4 and rate matched to transmission on $N_{re}^{PCC} = 98$ resource elements with $N_{bps} = 2$ for QPSK modulation and with single spatial stream $N_{SS} = 1$ as specified in clause 6.1.5. For all PCC transmissions redundancy version number $rv_{idx} = 0$.

The bits after channel encoding are denoted by $e_0, e_1, \ldots, e_{E-1}$, where $E = N_{re}^{PCC} N_{bps}$.

### 7.5.4 Scrambling

The block channel encoded bits $e_0, e_1, \ldots, e_{E-1}$, shall be scrambled, resulting in a block of scrambled bits:

$$d(i) = \big(e(i) + g(i)\big) \; mod \; 2$$

where the scrambling sequence g(i) is given by clause 6.2. The scrambling sequence shall be initialized with $g_{init} = 0x44454354$.

### 7.5.5 Symbol mapping

Bit sequence by $d_0, d, \ldots, d_{E-1}$ is mapped into complex valued QPSK modulation symbols $x_0, x_1, \ldots, x_{S-1}$ as defined in clause 6.3.1.2. The length of the modulated symbol vector is $S = N_{re}^{PCC}$.

### 7.5.6 Spatial multiplexing

Modulated symbols $x_0, x_1, \ldots, x_{S-1}$ are mapped to 0th spatial stream according to clause 6.3.2, thus:

$$x^{(0)}(i) = x(i)$$

where $i = 0, \ldots, S-1$ and $S = N_{re}^{PCC}$.

### 7.5.7 Transmit stream mapping

Modulated symbols $x^{(0)}(i)$ from spatial stream 0 in case of single antenna transmission or spatial multiplexing are mapped directly to corresponding transmit stream as specified in clause 6.3.3.1 or in case of transmit diversity precoded as specified in clause 6.3.3.2 producing output $y^{(t)}(i), i = 0, \ldots, S-1$, where $S = N_{re}^{PCC}$ and transmit streams $t = 0, \ldots, N_{TS} - 1$ according to the transmission mode.

### 7.5.8 Beamforming

The modulated symbols $y^{(t)}(i), i = 0, \ldots, S-1$ from transmit streams $t = 0, \ldots, N_{TS} - 1$ are beamformed according to the transmit mode as specified in clause 6.3.4 to produce PCC transmission $z^{(p)}(i), \; i = 0, \ldots, S$ on antenna port $p = 0, \ldots, N_{TX} - 1$.

### 7.5.9 Resource element mapping

Beamformed symbols symbols $z^{(p)}(i), \; i = 0, \ldots, N_{re}^{PCC} - 1, p = 0, \ldots, N_{TX} - 1$ are mapped into $N_{re}^{PCC} = 98$ subcarriers as defined in clause 5.2.4.

# 7.6 Physical Data Channel (PDC) encoding

## 7.6.1 Overall description

PDC encoding procedure is depicted in Figure 7.6.1-1.

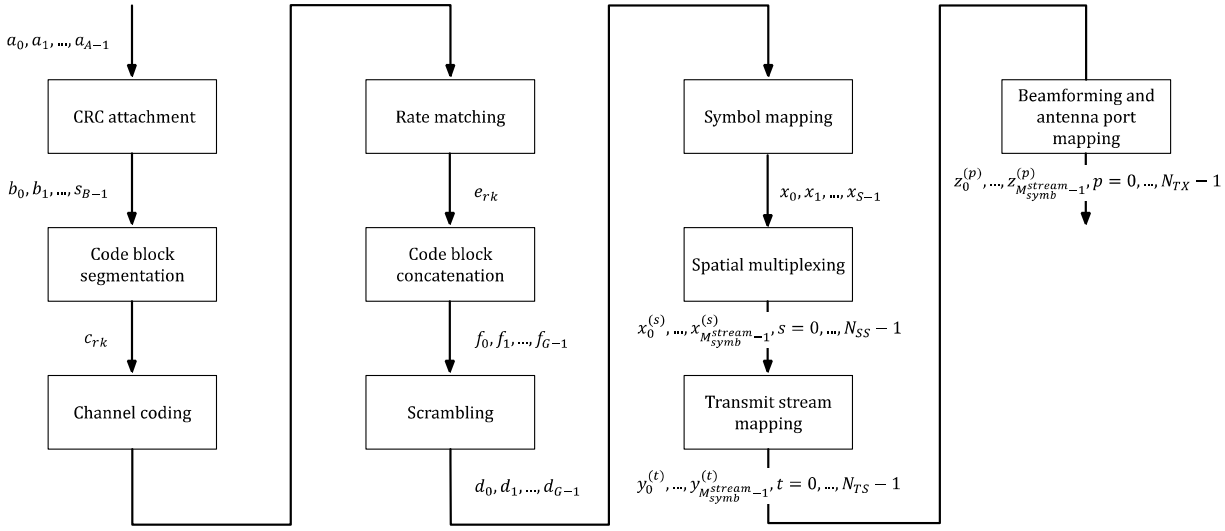



**Figure 7.6.1-1: Physical Data Channel Encoding**

## 7.6.2 CRC calculation

Physical data channel bits are $a_0, a_1, \cdots, a_{A-1}$, where the length of transport block is given by clause 5.3 according to the transmission allocation.

Parity bits for the transport block are generated by using the polynome $g_{CRC24A}(D)$ according to clause 6.1.2. The bits after CRC attachment are denoted by $b_0, b_1, \ldots, b_{B-1}$, where $B = A+ L$, where $L = 24$.

## 7.6.3 Code block segmentation

Transport block is segmented into *C* independently turbo encoded blocks, each individually protected with 24-bit CRC as specified in clause 6.1.3.

## 7.6.4 Channel coding & rate matching

Each of the code block segments are individually channel coded and rate matched as specified in clauses 6.1.4 and 6.1.5. All non-HARQ transmissions shall use redundancy version index $rv_{idx} = 0$ and set $N_{cb} = K_w$ in clause 6.1.5.3.

## 7.6.5 Code block concatenation

Code block segments are concatenated according to clause 6.1.6.

## 7.6.6 Scrambling

The channel encoded and concatenated codeblocks $f_0, f, \ldots, f_{G-1}$, where $G = N_{ss} \cdot N_{re}^{PDC} \cdot N_{bps}$ shall be scrambled resulting in a block of scrambled bits:

$$d(i) = \big(f(i) + g(i)\big) \, mod \; 2$$

where the scrambling sequence $g(i)$ is given by clause 6.2.

If transmission is using Physical Layer Control Field: Type 2 as specified in ETSI TS 103 636-4 [3], clause 6.2 the scrambling sequence shall be initialized with 24 MSB bits of the NetworkID ETSI TS 103 636-4 [3], clause 4.2.3:

$$g_{init} = (\text{Network ID} \gg 8)\ \&\ 0\text{x}00\text{ffffff}$$

If transmission is using Physical Layer Control Field: Type 1 as specified in ETSI TS 103 636-4 [3], clause 6.2 the scrambling sequence shall be initialized with 8 LSB bits of the Network ID ETSI TS 103 636-4 [3], clause 4.2.3:

$$g_{init} = \text{Network ID}\ \&\ 0\text{x}000000\text{ff}$$

## 7.6.7 Symbol mapping

Bit sequence by $d_0, d_1, \ldots, d_{G-1}$ is mapped into complex valued modulation symbols $x_0, x_1, \ldots, x_{S-1}$ as defined in clause 6.3.1 according to the MCS selected for the transport block. The length of the modulated symbol vector is $S = N_{re}^{PDC} \cdot N_{SS}$.

## 7.6.8 Spatial multiplexing

Modulated symbols $x_0, x_1, \ldots, x_{S-1}$ are mapped to spatial streams $x_i^{(0)}, \ldots, x_i^{(N_{SS}-1)}$ according to clause 6.3.2 for the selected transmission mode, where $S = N_{re}^{PDC} \cdot N_{SS}$ and $i = 0, \ldots, N_{re}^{PDC} - 1$.

## 7.6.9 Transmit stream mapping

Modulated symbols from spatial stream mapping $x_i^{(0)}, \ldots, x_i^{(N_{SS}-1)}$ are precoded to $y_i^{(0)}, \ldots, y_i^{(N_{TS}-1)}$ according to the transmission mode, where $N_{TS}$ is the number of transmit streams and $i = 0, \ldots, N_{re}^{PDC} - 1$.

For single antenna transmission the mapping is specified in clause 6.3.3.1.

For transmit diversity transmission is precoded is specified in clause 6.3.3.2.

For spatially multiplexed transmissions the mapping is specified in clause 6.3.3.1.

## 7.6.10 Beamforming and antenna port mapping

The modulated symbols $y_i^{(t)}, i = 0, \ldots, N_{re}^{PDC} - 1$ from transmit streams $t = 0, \ldots, N_{TS} - 1$ are beamformed according to the transmit mode as specified in clause 6.3.4 to produce PDC transmission $z_i^{(p)}, i = 0, \ldots, N_{re}^{PDC} - 1$ on antenna ports $p = 0, \ldots, N_{TX} - 1$.

## 7.6.11 Resource element mapping

Beamformed symbols $z_i^{(p)}, i = 0, \ldots, N_{re}^{PDC} - 1, p = 0, \ldots, N_{TX} - 1$ are mapped into $N_{re}^{PDC}$ subcarriers as defined in clause 5.2.5.

# Annex A (normative): Modulation and coding schemes

Modulation and coding schemes are listed in Table A-1, where $N_{bps}$ denotes the number of bits per modulation symbol and $R$ denotes the coding rate.

**Table A-1: Modulation and coding schemes**

| MCS Index | Modulation | $N_{bps}$ | $R$ |
|---|---|---|---|
| 0 | BPSK | 1 | 1/2 |
| 1 | QPSK | 2 | 1/2 |
| 2 | QPSK | 2 | 3/4 |
| 3 | 16-QAM | 4 | 1/2 |
| 4 | 16-QAM | 4 | 3/4 |
| 5 | 64-QAM | 6 | 2/3 |
| 6 | 64-QAM | 6 | 3/4 |
| 7 | 64-QAM | 6 | 5/6 |
| 8 | 256-QAM | 8 | 3/4 |
| 9 | 256-QAM | 8 | 5/6 |
| 10 | 1024-QAM | 10 | 3/4 |
| 11 | 1024-QAM | 10 | 5/6 |

# Annex B (normative): Radio Device Capabilities

## B.1 Introduction

Physical layer radio device capabilities to be signalled at connection setup between radio devices as defined in ETSI TS 103 636-4 [3] are defined here.

## B.2 Radio Device Capabilities

| | |
|---|---|
| Maximum number of spatial streams | Capability to receive single spatial stream, 2 spatial streams, 4 spatial streams or 8 spatial streams. Single spatial stream is the default configuration for all radio devices. See clause 7.2.<br>Larger spatial stream support implies support for smaller spatial stream support. |
| Reception for transmit diversity | Supports reception of 1 antenna, 2 antenna, 4 antenna or 8 antennas of TX diversity transmission. See clause 7.2.<br>Larger transmit antenna configuration implies support for the smaller antenna configurations. |
| Subcarrier width scaling - $\mu$ | Subcarrier widths of 27, 54, 108 and 216 kHz. See clause 4.3. |
| Fourier transform scaling - $\beta$ | Fourier transform size of 64, 128, 256, 512, 768 or 1 024. See clause 4.3. |
| Maximum modulation and coding scheme | Maximum supported modulation and coding scheme. See Annex A. RD shall support at least MCS 1. |
| Number of HARQ processes | Support for 1, 2, 4 or 8 HARQ processes. See clause 6.1.5. |
| Soft-buffer size | HARQ soft buffer size of at least $N_{soft}$ = 16 000, 25 344, 32 000, 64 000, 128 000, 256 000, 512 000, 1 024 000, 2 048 000 bytes. See clause 6.1.5. |
| Codeblock segment size | Maximum turbo codeblock segment size is fixed to $Z$ = 2 048 and need not be signalled. See clause 6.1.3. |

# Annex C (informative): Examples of transport block sizes and maximum achievable data rates

## C.1 Introduction

In this annex just a few examples of Transport Block Sizes (TBSs) and achievable peak datarates are listed. Short transmission packet suffers relatively more of synchronization, channel training and control channel overheads compared to long packets which can be seen by comparing examples of clauses C.2 to C.4, which list TBS lengths and peak datarates for single slot, dual slot and quad slot transmissions. The transport block size depends on:

1) transmission length which can range from single subslot up to sixteen slot transmission with subslot (5 OFDM symbol) granularity;

2) modulation and coding scheme;

3) number of transmission streams; and

4) system bandwidth.

Number of unique transmission configurations is high. Therefore, listing them all here is impractical. Use clause 5.3 to derive TBS for transmission configuration and calculate achievable data rate using transmission duration and transmission duty-cycle.

# C.2 Single slot transmission, single spatial stream

**Table C.2-1: Transport block sizes (in bits) for single slot transmission and single spatial stream with maximum turbo code block size of 2 048 bits**

TRANSPORT BLOCK SIZES

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 136 | 296 | 456 | 616 | 936 | 1 256 | 1 416 | 1 576 | 1 896 | 2 040 | 2 296 | 2 552 |
| | 2 | 344 | 712 | 1 064 | 1 448 | 2 104 | 2 872 | 3 256 | 3 576 | 4 320 | 4 832 | 5 408 | 6 048 |
| | 4 | 760 | 1 544 | 2 296 | 3 064 | 4 640 | 6 152 | 6 984 | 7 752 | 9 328 | 10 328 | 11 672 | 12 928 |
| | 8 | 1 576 | 3 128 | 4 768 | 6 344 | 9 584 | 12 800 | 14 440 | 16 040 | 19 256 | 21 408 | 24 136 | 26 776 |
| | 12 | 2 360 | 4 832 | 7 240 | 9 712 | 14 568 | 19 448 | 21 920 | 24 368 | 29 248 | 32 464 | 36 576 | 40 624 |
| | 16 | 3 192 | 6 472 | 9 776 | 12 992 | 19 576 | 26 096 | 29 376 | 32 656 | 39 176 | 43 544 | 49 040 | 54 496 |

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 1 | 368 | 760 | 1 160 | 1 544 | 2 296 | 3 064 | 3 512 | 3 896 | 4 640 | 5 216 | 5 856 | 6 472 |
| | 2 | 808 | 1 640 | 2 424 | 3 256 | 4 960 | 6 600 | 7 496 | 8 304 | 9 968 | 11 096 | 12 480 | 13 888 |
| | 4 | 1 704 | 3 384 | 5 088 | 6 792 | 10 264 | 13 696 | 15 464 | 17 168 | 20 576 | 22 920 | 25 776 | 28 608 |
| | 8 | 3 448 | 6 920 | 10 392 | 13 888 | 20 896 | 27 864 | 31 400 | 34 872 | 41 840 | 46 528 | 52 344 | 58 160 |
| | 12 | 5 216 | 10 456 | 15 720 | 20 960 | 31 528 | 42 032 | 47 336 | 52 600 | 63 080 | 70 112 | 78 912 | 87 688 |
| | 16 | 6 984 | 14 016 | 21 024 | 28 056 | 42 160 | 56 200 | 63 272 | 70 304 | 84 344 | 93 760 | 105 456 | 117 176 |

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 1 | 904 | 1 832 | 2 680 | 3 640 | 5 472 | 7 304 | 8 176 | 9 136 | 10 968 | 12 160 | 13 760 | 15 272 |
| | 2 | 1 864 | 3 704 | 5 600 | 7 496 | 11 288 | 15 016 | 16 912 | 18 808 | 22 600 | 25 136 | 28 248 | 31 400 |
| | 4 | 3 768 | 7 560 | 11 416 | 15 208 | 22 856 | 30 504 | 34 384 | 38 176 | 45 824 | 50 936 | 57 328 | 63 720 |
| | 8 | 7 624 | 15 336 | 23 048 | 30 696 | 46 144 | 61 504 | 69 216 | 76 928 | 92 312 | 102 624 | 115 448 | 128 232 |
| | 12 | 11 480 | 23 048 | 34 616 | 46 208 | 69 344 | 92 504 | 104 072 | 115 640 | 138 800 | 154 224 | 173 528 | 192 808 |
| | 16 | 15 400 | 30 824 | 46 272 | 61 696 | 92 632 | 123 480 | 138 992 | 154 416 | 185 288 | 205 912 | 231 648 | 257 384 |

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 8 | 1 | 1 928 | 3 832 | 5 792 | 7 688 | 11 608 | 15 464 | 17 424 | 19 384 | 23 240 | 25 840 | 29 120 | 32 360 |
| | 2 | 3 896 | 7 816 | 11 736 | 15 656 | 23 560 | 31 400 | 35 384 | 39 304 | 47 144 | 52 408 | 58 968 | 65 552 |
| | 4 | 7 880 | 15 784 | 23 688 | 31 592 | 47 464 | 63 272 | 71 240 | 79 144 | 94 976 | 105 520 | 118 752 | 131 960 |
| | 8 | 15 848 | 31 720 | 47 592 | 63 464 | 95 272 | 127 040 | 142 976 | 158 848 | 190 592 | 211 792 | 238 296 | 264 800 |
| | 12 | 23 816 | 47 656 | 71 496 | 95 336 | 143 104 | 190 784 | 214 688 | 238 552 | 286 232 | 318 040 | 357 840 | 397 616 |
| | 16 | 31 784 | 63 592 | 95 400 | 127 232 | 190 912 | 254 552 | 286 424 | 318 232 | 381 872 | 424 312 | 477 384 | 530 432 |

**Table C.2-2: Single channel maximum throughput for single slot transmission with single spatial stream with maximum turbo code block size of 2 048 bits**

DATARATE (Mbps)

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0,326 | 0,710 | 1,094 | 1,478 | 2,246 | 3,014 | 3,398 | 3,782 | 4,550 | 4,896 | 5,510 | 6,125 |
| | 2 | 0,826 | 1,709 | 2,554 | 3,475 | 5,050 | 6,893 | 7,814 | 8,582 | 10,368 | 11,597 | 12,979 | 14,515 |
| | 4 | 1,824 | 3,706 | 5,510 | 7,354 | 11,136 | 14,765 | 16,762 | 18,605 | 22,387 | 24,787 | 28,013 | 31,027 |
| | 8 | 3,782 | 7,507 | 11,443 | 15,226 | 23,002 | 30,720 | 34,656 | 38,496 | 46,214 | 51,379 | 57,926 | 64,262 |
| | 12 | 5,664 | 11,597 | 17,376 | 23,309 | 34,963 | 46,675 | 52,608 | 58,483 | 70,195 | 77,914 | 87,782 | 97,498 |
| | 16 | 7,661 | 15,533 | 23,462 | 31,181 | 46,982 | 62,630 | 70,502 | 78,374 | 94,022 | 104,506 | 117,696 | 130,790 |

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 1 | 0,883 | 1,824 | 2,784 | 3,706 | 5,510 | 7,354 | 8,429 | 9,350 | 11,136 | 12,518 | 14,054 | 15,533 |
| | 2 | 1,939 | 3,936 | 5,818 | 7,814 | 11,904 | 15,840 | 17,990 | 19,930 | 23,923 | 26,630 | 29,952 | 33,331 |
| | 4 | 4,090 | 8,122 | 12,211 | 16,301 | 24,634 | 32,870 | 37,114 | 41,203 | 49,382 | 55,008 | 61,862 | 68,659 |
| | 8 | 8,275 | 16,608 | 24,941 | 33,331 | 50,150 | 66,874 | 75,360 | 83,693 | 100,416 | 111,667 | 125,626 | 139,584 |
| | 12 | 12,518 | 25,094 | 37,728 | 50,304 | 75,667 | 100,877 | 113,606 | 126,240 | 151,392 | 168,269 | 189,389 | 210,451 |
| | 16 | 16,762 | 33,638 | 50,458 | 67,334 | 101,184 | 134,880 | 151,853 | 168,730 | 202,426 | 225,024 | 253,094 | 281,222 |

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 1 | 2,170 | 4,397 | 6,432 | 8,736 | 13,133 | 17,530 | 19,622 | 21,926 | 26,323 | 29,184 | 33,024 | 36,653 |
| | 2 | 4,474 | 8,890 | 13,440 | 17,990 | 27,091 | 36,038 | 40,589 | 45,139 | 54,240 | 60,326 | 67,795 | 75,360 |
| | 4 | 9,043 | 18,144 | 27,398 | 36,499 | 54,854 | 73,210 | 82,522 | 91,622 | 109,978 | 122,246 | 137,587 | 152,928 |
| | 8 | 18,298 | 36,806 | 55,315 | 73,670 | 110,746 | 147,610 | 166,118 | 184,627 | 221,549 | 246,298 | 277,075 | 307,757 |
| | 12 | 27,552 | 55,315 | 83,078 | 110,899 | 166,426 | 222,010 | 249,773 | 277,536 | 333,120 | 370,138 | 416,467 | 462,739 |
| | 16 | 36,960 | 73,978 | 111,053 | 148,070 | 222,317 | 296,352 | 333,581 | 370,598 | 444,691 | 494,189 | 555,955 | 617,722 |

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 8 | 1 | 4,627 | 9,197 | 13,901 | 18,451 | 27,859 | 37,114 | 41,818 | 46,522 | 55,776 | 62,016 | 69,888 | 77,664 |
| | 2 | 9,350 | 18,758 | 28,166 | 37,574 | 56,544 | 75,360 | 84,922 | 94,330 | 113,146 | 125,779 | 141,523 | 157,325 |
| | 4 | 18,912 | 37,882 | 56,851 | 75,821 | 113,914 | 151,853 | 170,976 | 189,946 | 227,942 | 253,248 | 285,005 | 316,704 |
| | 8 | 38,035 | 76,128 | 114,221 | 152,314 | 228,653 | 304,896 | 343,142 | 381,235 | 457,421 | 508,301 | 571,910 | 635,520 |
| | 12 | 57,158 | 114,374 | 171,590 | 228,806 | 343,450 | 457,882 | 515,251 | 572,525 | 686,957 | 763,296 | 858,816 | 954,278 |
| | 16 | 76,282 | 152,621 | 228,960 | 305,357 | 458,189 | 610,925 | 687,418 | 763,757 | 916,493 | 1 018,349 | 1 145,722 | 1 273,037 |

# C.3 Dual slot transmission, single spatial stream

**Table C.3-1: Transport block sizes (in bits) for dual slot transmission and single spatial stream with maximum turbo code block size of 2 048 bits**

TRANSPORT BLOCK SIZES

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 400 | 824 | 1 256 | 1 672 | 2 488 | 3 320 | 3 768 | 4 128 | 5 024 | 5 536 | 6 280 | 6 984 |
| | 2 | 872 | 1 768 | 2 616 | 3 512 | 5 280 | 7 048 | 7 944 | 8 880 | 10 648 | 11 864 | 13 312 | 14 824 |
| | 4 | 1 800 | 3 576 | 5 408 | 7 240 | 10 904 | 14 568 | 16 400 | 18 232 | 21 920 | 24 368 | 27 416 | 30 504 |
| | 8 | 3 640 | 7 368 | 11 096 | 14 760 | 22 240 | 29 632 | 33 360 | 37 088 | 44 504 | 49 488 | 55 688 | 61 824 |
| | 12 | 5 536 | 11 096 | 16 720 | 22 280 | 33 488 | 44 672 | 50 320 | 55 880 | 67 064 | 74 544 | 83 896 | 93 184 |
| | 16 | 7 432 | 14 888 | 22 344 | 29 824 | 44 800 | 59 736 | 67 256 | 74 736 | 89 648 | 99 640 | 112 104 | 124 568 |

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 1 | 904 | 1 832 | 2 680 | 3 640 | 5 472 | 7 304 | 8 176 | 9 136 | 10 968 | 12 160 | 13 760 | 15 272 |
| | 2 | 1 864 | 3 704 | 5 600 | 7 496 | 11 288 | 15 016 | 16 912 | 18 808 | 22 600 | 25 136 | 28 248 | 31 400 |
| | 4 | 3 768 | 7 560 | 11 416 | 15 208 | 22 856 | 30 504 | 34 384 | 38 176 | 45 824 | 50 936 | 57 328 | 63 720 |
| | 8 | 7 624 | 15 336 | 23 048 | 30 696 | 46 144 | 61 504 | 69 216 | 76 928 | 92 312 | 102 624 | 115 448 | 128 232 |
| | 12 | 11 480 | 23 048 | 34 616 | 46 208 | 69 344 | 92 504 | 104 072 | 115 640 | 138 800 | 154 224 | 173 528 | 192 808 |
| | 16 | 15 400 | 30 824 | 46 272 | 61 696 | 92 632 | 123 480 | 138 992 | 154 416 | 185 288 | 205 912 | 231 648 | 257 384 |

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 1 | 1 960 | 3 896 | 5 856 | 7 816 | 11 800 | 15 720 | 17 680 | 19 640 | 23 624 | 26 224 | 29 504 | 32 784 |
| | 2 | 3 960 | 7 944 | 11 928 | 15 912 | 23 880 | 31 848 | 35 832 | 39 880 | 47 848 | 53 152 | 59 800 | 66 448 |
| | 4 | 7 944 | 15 976 | 24 008 | 32 040 | 48 104 | 64 168 | 72 200 | 80 232 | 96 296 | 107 056 | 120 392 | 133 792 |
| | 8 | 16 040 | 32 168 | 48 296 | 64 360 | 96 616 | 128 808 | 144 936 | 161 064 | 193 256 | 214 752 | 241 640 | 268 464 |
| | 12 | 24 136 | 48 296 | 72 520 | 96 680 | 145 064 | 193 448 | 217 672 | 241 832 | 290 216 | 322 472 | 362 824 | 403 112 |
| | 16 | 32 232 | 64 488 | 96 744 | 129 000 | 193 576 | 258 088 | 290 408 | 322 664 | 387 176 | 430 256 | 484 008 | 537 784 |

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 8 | 1 | 3 960 | 8 008 | 12 056 | 16 104 | 24 200 | 32 296 | 36 344 | 40 392 | 48 488 | 53 920 | 60 632 | 67 384 |
| | 2 | 8 072 | 16 208 | 24 368 | 32 464 | 48 784 | 65 040 | 73 200 | 81 360 | 97 616 | 108 504 | 122 096 | 135 624 |
| | 4 | 16 272 | 32 592 | 48 912 | 65 232 | 97 936 | 130 576 | 146 960 | 163 280 | 195 920 | 217 736 | 244 920 | 272 128 |
| | 8 | 32 656 | 65 360 | 98 064 | 130 768 | 196 240 | 261 624 | 294 392 | 327 096 | 392 504 | 436 136 | 490 656 | 545 176 |
| | 12 | 49 040 | 98 128 | 147 216 | 196 304 | 294 520 | 392 672 | 441 824 | 490 912 | 589 064 | 654 536 | 736 392 | 818 224 |
| | 16 | 65 424 | 130 896 | 196 344 | 261 816 | 392 800 | 523 744 | 589 256 | 654 728 | 785 648 | 873 000 | 982 104 | 1 091 208 |

**Table C.3-2: Single channel maximum throughput for dual slot transmission with single spatial stream with maximum turbo code block size of 2 048 bits**

DATARATE (Mbps)

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0,480 | 0,989 | 1,507 | 2,006 | 2,986 | 3,984 | 4,522 | 4,954 | 6,029 | 6,643 | 7,536 | 8,381 |
| | 2 | 1,046 | 2,122 | 3,139 | 4,214 | 6,336 | 8,458 | 9,533 | 10,656 | 12,778 | 14,237 | 15,974 | 17,789 |
| | 4 | 2,160 | 4,291 | 6,490 | 8,688 | 13,085 | 17,482 | 19,680 | 21,878 | 26,304 | 29,242 | 32,899 | 36,605 |
| | 8 | 4,368 | 8,842 | 13,315 | 17,712 | 26,688 | 35,558 | 40,032 | 44,506 | 53,405 | 59,386 | 66,826 | 74,189 |
| | 12 | 6,643 | 13,315 | 20,064 | 26,736 | 40,186 | 53,606 | 60,384 | 67,056 | 80,477 | 89,453 | 100,675 | 111,821 |
| | 16 | 8,918 | 17,866 | 26,813 | 35,789 | 53,760 | 71,683 | 80,707 | 89,683 | 107,578 | 119,568 | 134,525 | 149,482 |

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 1 | 1,085 | 2,198 | 3,216 | 4,368 | 6,566 | 8,765 | 9,811 | 10,963 | 13,162 | 14,592 | 16,512 | 18,326 |
| | 2 | 2,237 | 4,445 | 6,720 | 8,995 | 13,546 | 18,019 | 20,294 | 22,570 | 27,120 | 30,163 | 33,898 | 37,680 |
| | 4 | 4,522 | 9,072 | 13,699 | 18,250 | 27,427 | 36,605 | 41,261 | 45,811 | 54,989 | 61,123 | 68,794 | 76,464 |
| | 8 | 9,149 | 18,403 | 27,658 | 36,835 | 55,373 | 73,805 | 83,059 | 92,314 | 110,774 | 123,149 | 138,538 | 153,878 |
| | 12 | 13,776 | 27,658 | 41,539 | 55,450 | 83,213 | 111,005 | 124,886 | 138,768 | 166,560 | 185,069 | 208,234 | 231,370 |
| | 16 | 18,480 | 36,989 | 55,526 | 74,035 | 111,158 | 148,176 | 166,790 | 185,299 | 222,346 | 247,094 | 277,978 | 308,861 |

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 1 | 2,352 | 4,675 | 7,027 | 9,379 | 14,160 | 18,864 | 21,216 | 23,568 | 28,349 | 31,469 | 35,405 | 39,341 |
| | 2 | 4,752 | 9,533 | 14,314 | 19,094 | 28,656 | 38,218 | 42,998 | 47,856 | 57,418 | 63,782 | 71,760 | 79,738 |
| | 4 | 9,533 | 19,171 | 28,810 | 38,448 | 57,725 | 77,002 | 86,640 | 96,278 | 115,555 | 128,467 | 144,470 | 160,550 |
| | 8 | 19,248 | 38,602 | 57,955 | 77,232 | 115,939 | 154,570 | 173,923 | 193,277 | 231,907 | 257,702 | 289,968 | 322,157 |
| | 12 | 28,963 | 57,955 | 87,024 | 116,016 | 174,077 | 232,138 | 261,206 | 290,198 | 348,259 | 386,966 | 435,389 | 483,734 |
| | 16 | 38,678 | 77,386 | 116,093 | 154,800 | 232,291 | 309,706 | 348,490 | 387,197 | 464,611 | 516,307 | 580,810 | 645,341 |

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 8 | 1 | 4,752 | 9,610 | 14,467 | 19,325 | 29,040 | 38,755 | 43,613 | 48,470 | 58,186 | 64,704 | 72,758 | 80,861 |
| | 2 | 9,686 | 19,450 | 29,242 | 38,957 | 58,541 | 78,048 | 87,840 | 97,632 | 117,139 | 130,205 | 146,515 | 162,749 |
| | 4 | 19,526 | 39,110 | 58,694 | 78,278 | 117,523 | 156,691 | 176,352 | 195,936 | 235,104 | 261,283 | 293,904 | 326,554 |
| | 8 | 39,187 | 78,432 | 117,677 | 156,922 | 235,488 | 313,949 | 353,270 | 392,515 | 471,005 | 523,363 | 588,787 | 654,211 |
| | 12 | 58,848 | 117,754 | 176,659 | 235,565 | 353,424 | 471,206 | 530,189 | 589,094 | 706,877 | 785,443 | 883,670 | 981,869 |
| | 16 | 78,509 | 157,075 | 235,613 | 314,179 | 471,360 | 628,493 | 707,107 | 785,674 | 942,778 | 1 047,600 | 1 178,525 | 1 309,450 |

# C.4 Quad slot transmission, single spatial stream

**Table C.4-1: Transport block sizes (in bits) for quad slot transmission and single spatial stream with maximum turbo code block size of 2 048 bits**

TRANSPORT BLOCK SIZES

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 920 | 1 864 | 2 744 | 3 704 | 5 600 | 7 496 | 8 432 | 9 392 | 11 288 | 12 544 | 14 144 | 15 720 |
| | 2 | 1 928 | 3 832 | 5 792 | 7 688 | 11 608 | 15 464 | 17 424 | 19 384 | 23 240 | 25 840 | 29 120 | 32 360 |
| | 4 | 3 896 | 7 816 | 11 736 | 15 656 | 23 560 | 31 400 | 35 384 | 39 304 | 47 144 | 52 408 | 58 968 | 65 552 |
| | 8 | 7 880 | 15 784 | 23 688 | 31 592 | 47 464 | 63 272 | 71 240 | 79 144 | 94 976 | 105 520 | 118 752 | 131 960 |
| | 12 | 11 864 | 23 752 | 35 640 | 47 528 | 71 368 | 95 144 | 107 120 | 119 008 | 142 784 | 158 656 | 178 512 | 198 368 |
| | 16 | 15 848 | 31 720 | 47 592 | 63 464 | 95 272 | 127 040 | 142 976 | 158 848 | 190 592 | 211 792 | 238 296 | 264 800 |

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 1 | 1 960 | 3 896 | 5 856 | 7 816 | 11 800 | 15 720 | 17 680 | 19 640 | 23 624 | 26 224 | 29 504 | 32 784 |
| | 2 | 3 960 | 7 944 | 11 928 | 15 912 | 23 880 | 31 848 | 35 832 | 39 880 | 47 848 | 53 152 | 59 800 | 66 448 |
| | 4 | 7 944 | 15 976 | 24 008 | 32 040 | 48 104 | 64 168 | 72 200 | 80 232 | 96 296 | 107 056 | 120 392 | 133 792 |
| | 8 | 16 040 | 32 168 | 48 296 | 64 360 | 96 616 | 128 808 | 144 936 | 161 064 | 193 256 | 214 752 | 241 640 | 268 464 |
| | 12 | 24 136 | 48 296 | 72 520 | 96 680 | 145 064 | 193 448 | 217 672 | 241 832 | 290 216 | 322 472 | 362 824 | 403 112 |
| | 16 | 32 232 | 64 488 | 96 744 | 129 000 | 193 576 | 258 088 | 290 408 | 322 664 | 387 176 | 430 256 | 484 008 | 537 784 |

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 1 | 4 024 | 8 072 | 12 160 | 16 208 | 24 368 | 32 528 | 36 576 | 40 688 | 48 848 | 54 240 | 61 056 | 67 832 |
| | 2 | 8 112 | 16 336 | 24 496 | 32 720 | 49 104 | 65 488 | 73 712 | 81 936 | 98 320 | 109 272 | 122 928 | 136 584 |
| | 4 | 16 336 | 32 784 | 49 232 | 65 680 | 98 576 | 131 472 | 147 896 | 164 344 | 197 240 | 219 184 | 246 584 | 274 024 |
| | 8 | 32 848 | 65 808 | 98 768 | 131 640 | 197 560 | 263 392 | 296 352 | 329 312 | 395 144 | 439 120 | 494 000 | 548 840 |
| | 12 | 49 360 | 98 768 | 148 216 | 197 624 | 296 480 | 395 336 | 444 808 | 494 192 | 593 048 | 658 968 | 741 376 | 823 744 |
| | 16 | 65 872 | 131 768 | 197 688 | 263 584 | 395 464 | 527 280 | 593 240 | 659 160 | 790 976 | 878 880 | 988 752 | 1 098 624 |

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 8 | 1 | 8 176 | 16 400 | 24 688 | 32 912 | 49 424 | 65 936 | 74 224 | 82 448 | 98 960 | 109 952 | 123 736 | 137 480 |
| | 2 | 16 464 | 33 040 | 49 616 | 66 128 | 99 256 | 132 344 | 148 920 | 165 496 | 198 560 | 220 632 | 248 288 | 275 856 |
| | 4 | 33 104 | 66 256 | 99 384 | 132 536 | 198 880 | 265 160 | 298 376 | 331 528 | 397 808 | 442 016 | 497 304 | 552 568 |
| | 8 | 66 320 | 132 664 | 199 008 | 265 352 | 398 128 | 530 816 | 597 224 | 663 592 | 796 280 | 884 760 | 995 400 | 1 106 016 |
| | 12 | 99 512 | 199 072 | 298 632 | 398 192 | 597 352 | 796 472 | 896 096 | 995 656 | 1 194 752 | 1 327 528 | 1 493 496 | 1 659 464 |
| | 16 | 132 728 | 265 480 | 398 256 | 531 008 | 796 600 | 1 062 128 | 1 194 944 | 1 327 720 | 1 593 224 | 1 770 272 | 1 991 592 | 2 212 888 |

**Table C.4-2: Single channel maximum throughput for quad slot transmission with single spatial stream with maximum turbo code block size of 2 048 bits**

DATARATE (Mbps)

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0,552 | 1,118 | 1,646 | 2,222 | 3,360 | 4,498 | 5,059 | 5,635 | 6,773 | 7,526 | 8,486 | 9,432 |
| | 2 | 1,157 | 2,299 | 3,475 | 4,613 | 6,965 | 9,278 | 10,454 | 11,630 | 13,944 | 15,504 | 17,472 | 19,416 |
| | 4 | 2,338 | 4,690 | 7,042 | 9,394 | 14,136 | 18,840 | 21,230 | 23,582 | 28,286 | 31,445 | 35,381 | 39,331 |
| | 8 | 4,728 | 9,470 | 14,213 | 18,955 | 28,478 | 37,963 | 42,744 | 47,486 | 56,986 | 63,312 | 71,251 | 79,176 |
| | 12 | 7,118 | 14,251 | 21,384 | 28,517 | 42,821 | 57,086 | 64,272 | 71,405 | 85,670 | 95,194 | 107,107 | 119,021 |
| | 16 | 9,509 | 19,032 | 28,555 | 38,078 | 57,163 | 76,224 | 85,786 | 95,309 | 114,355 | 127,075 | 142,978 | 158,880 |

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 1 | 1,176 | 2,338 | 3,514 | 4,690 | 7,080 | 9,432 | 10,608 | 11,784 | 14,174 | 15,734 | 17,702 | 19,670 |
| | 2 | 2,376 | 4,766 | 7,157 | 9,547 | 14,328 | 19,109 | 21,499 | 23,928 | 28,709 | 31,891 | 35,880 | 39,869 |
| | 4 | 4,766 | 9,586 | 14,405 | 19,224 | 28,862 | 38,501 | 43,320 | 48,139 | 57,778 | 64,234 | 72,235 | 80,275 |
| | 8 | 9,624 | 19,301 | 28,978 | 38,616 | 57,970 | 77,285 | 86,962 | 96,638 | 115,954 | 128,851 | 144,984 | 161,078 |
| | 12 | 14,482 | 28,978 | 43,512 | 58,008 | 87,038 | 116,069 | 130,603 | 145,099 | 174,130 | 193,483 | 217,694 | 241,867 |
| | 16 | 19,339 | 38,693 | 58,046 | 77,400 | 116,146 | 154,853 | 174,245 | 193,598 | 232,306 | 258,154 | 290,405 | 322,670 |

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 1 | 2,414 | 4,843 | 7,296 | 9,725 | 14,621 | 19,517 | 21,946 | 24,413 | 29,309 | 32,544 | 36,634 | 40,699 |
| | 2 | 4,867 | 9,802 | 14,698 | 19,632 | 29,462 | 39,293 | 44,227 | 49,162 | 58,992 | 65,563 | 73,757 | 81,950 |
| | 4 | 9,802 | 19,670 | 29,539 | 39,408 | 59,146 | 78,883 | 88,738 | 98,606 | 118,344 | 131,510 | 147,950 | 164,414 |
| | 8 | 19,709 | 39,485 | 59,261 | 78,984 | 118,536 | 158,035 | 177,811 | 197,587 | 237,086 | 263,472 | 296,400 | 329,304 |
| | 12 | 29,616 | 59,261 | 88,930 | 118,574 | 177,888 | 237,202 | 266,885 | 296,515 | 355,829 | 395,381 | 444,826 | 494,246 |
| | 16 | 39,523 | 79,061 | 118,613 | 158,150 | 237,278 | 316,368 | 355,944 | 395,496 | 474,586 | 527,328 | 593,251 | 659,174 |

| μ | β \ MCS | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 8 | 1 | 4,906 | 9,840 | 14,813 | 19,747 | 29,654 | 39,562 | 44,534 | 49,469 | 59,376 | 65,971 | 74,242 | 82,488 |
| | 2 | 9,878 | 19,824 | 29,770 | 39,677 | 59,554 | 79,406 | 89,352 | 99,298 | 119,136 | 132,379 | 148,973 | 165,514 |
| | 4 | 19,862 | 39,754 | 59,630 | 79,522 | 119,328 | 159,096 | 179,026 | 198,917 | 238,685 | 265,210 | 298,382 | 331,541 |
| | 8 | 39,792 | 79,598 | 119,405 | 159,211 | 238,877 | 318,490 | 358,334 | 398,155 | 477,768 | 530,856 | 597,240 | 663,610 |
| | 12 | 59,707 | 119,443 | 179,179 | 238,915 | 358,411 | 477,883 | 537,658 | 597,394 | 716,851 | 796,517 | 896,098 | 995,678 |
| | 16 | 79,637 | 159,288 | 238,954 | 318,605 | 477,960 | 637,277 | 716,966 | 796,632 | 955,934 | 1 062,163 | 1 194,955 | 1 327,733 |

# Annex D (informative): Change History

| Date | Version | Information about changes |
|---|---|---|
| July 2020 | 1.1.1 | First publication of the TS |
| October 2020 | 1.1.2 | Implemented Change Requests:<br>DECT(20)000309r1 DRS correction<br>DECT(20)000310r1 Precoding correction<br>DECT(20)000311r1 Scrambling correction<br>DECT(20)000312r1 STF correction<br>These CRs were approved DECT-Telco 2.11.2020<br><br>Version 1.1.2 prepared by the Rapporteur |
| November 2020 | 1.1.3 | Implemented Change Requests:<br>DECT(20)000332 Transmission Bandwidth Correction<br>This CR was approved in DECT-Telco 16.11.2020<br><br>Version 1.1.3 prepared by the Rapporteur |
| March 2021 | 1.1.4 | Implemented Change Requests:<br>DECT(21)000063 ETSI_TS_103_636-3_Physical_Layer_Editorial_Corrections<br>This CR was approved in DECT#89 Plenary<br><br>Version 1.1.4 prepared by the Rapporteur |
| April 2021 | 1.2.1 | Publication of the TS version 1.2.1 |
| November 2021 | 1.2.2 | Implemented Change Requests:<br><br>DECT(21)000305<br>These CRs were approved DECT#92-Plenary<br><br>Version 1.2.2 prepared by the Rapporteur |
| November 2021 | 1.2.3 | Added reference to Recommendation ITU-R M.2150 and description to Foreword<br><br>Version 1.2.3 prepared by the Rapporteur |
| November 2021 | 1.2.4 | Corrected informative reference<br><br>Version 1.2.4 prepared by the Rapporteur |
| June 2021 | 1.3.2 | Implemented Change Requests:<br>DECT(22)000121 – Editorial changes<br><br>Version 1.3.2 prepared by the Rapporteur |
| November 2022 | 1.3.3 | Implemented Change Requests:<br>DECT(22)095018 – Decreased PAPR STFs<br><br>Version 1.3.3 prepared by the Rapporteur |
| December 2022 | 1.3.4 | Implemented Change Request:<br>DECT(22)000287 – Draft ETSI TS 103 636-3 v1.3.4, physical layer capability signalling<br><br>Version 1.3.4 prepared by the Rapporteur |
| March 2023 | 1.4.2 | Implemented Change Request:<br>DECT(23)000017 – DRS sign fix<br><br>Version 1.4.2 prepared by the Rapporteur |
| June 2023 | 1.4.3 | Implemented Change Request:<br>DECT(23)000096_Correction_in_figure_4_4-1_and_other_corrections_in_part_3<br>DECT(23)000136 – PHY_pseudorandom_code_editorial_fix<br>DECT(23)000104r1 – Synchronization Training Field Cover Sequence<br>DECT(23)000133r1 – STF indexing fix<br>DECT(23)000135 – More corrections to part 3<br><br>Version 1.4.3 prepared by the Rapporteur |

| Date | Version | Information about changes |
|---|---|---|
| January 2024 | 2.0.1 | Implemented Change Request:<br>DECT(24)000015_Correction_to_TS_103_636_-_3__Rel2_<br>DECT(24)000024r1_For_transmitter_conformance_testing_transmission_without_STF<br><br>Version 2.0.1 prepared by the Rapporteur |
| April 2024 | 2.0.2 | Implemented Change Requests:<br>DECT(24)000078_cleanup_references_to_STF_use_for_MIMO.docx<br>DECT(24)000112_CR_on_RD_capabilities_for_release_1.docx<br><br>Version 2.0.2 prepared by the Rapporteur |
| September 2024 | 2.0.3 | Implemented Change Requests:<br>DECT(24)000237_Transmission_without_PDC.docx |

# History

| Version | Date | Status |
|---|---|---|
| V2.1.1 | October 2024 | Publication |
| V2.2.1 | May 2026 | Publication |
| | | |
| | | |